

Pankaj Jalluri

Engineering Portfolio

Mechanical Design • Product Development • Robotics



Button Cycle Tester

01

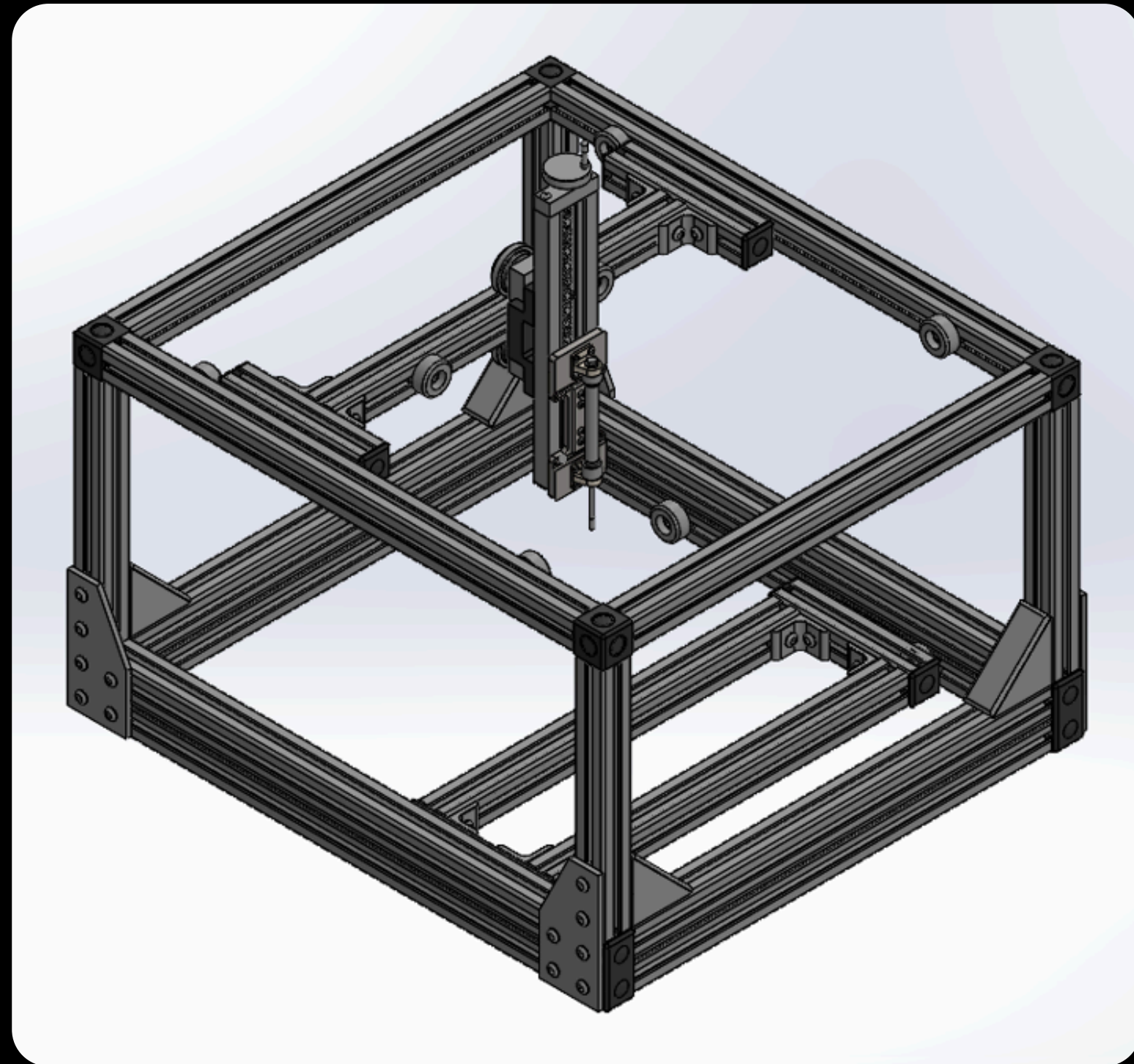
Overview

Designed an automated fixture to perform button cycle testing, repeatedly actuating product buttons (~3000 presses for each button) to evaluate durability and functionality over extended use.

Developed as an adaptable NPD verification platform for multiple ALTAIR portable gas detection products with varying sizes and button layouts.

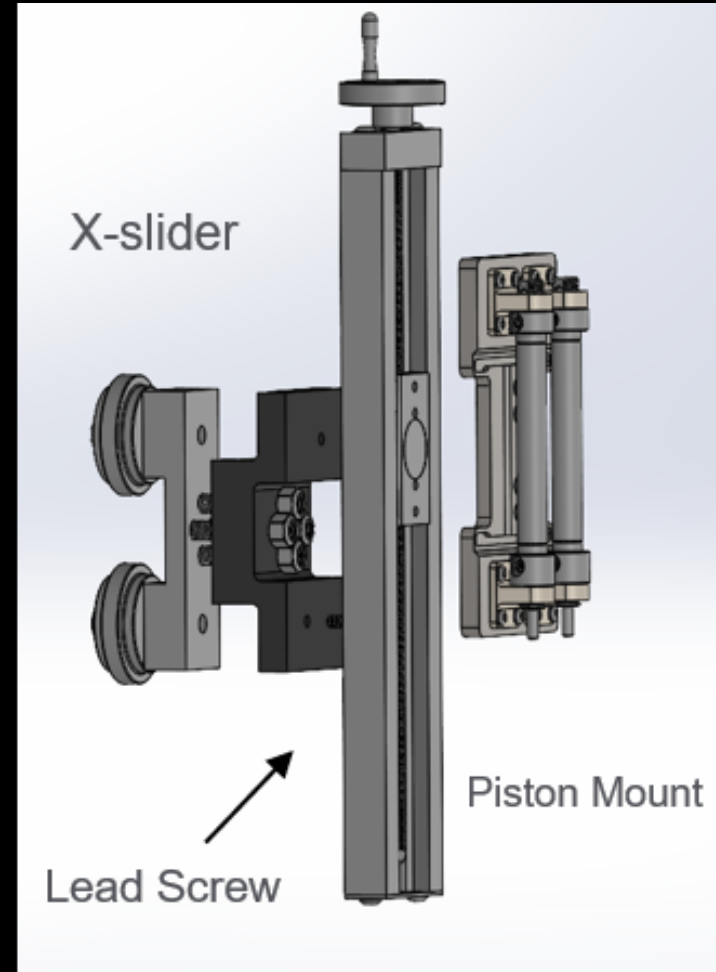
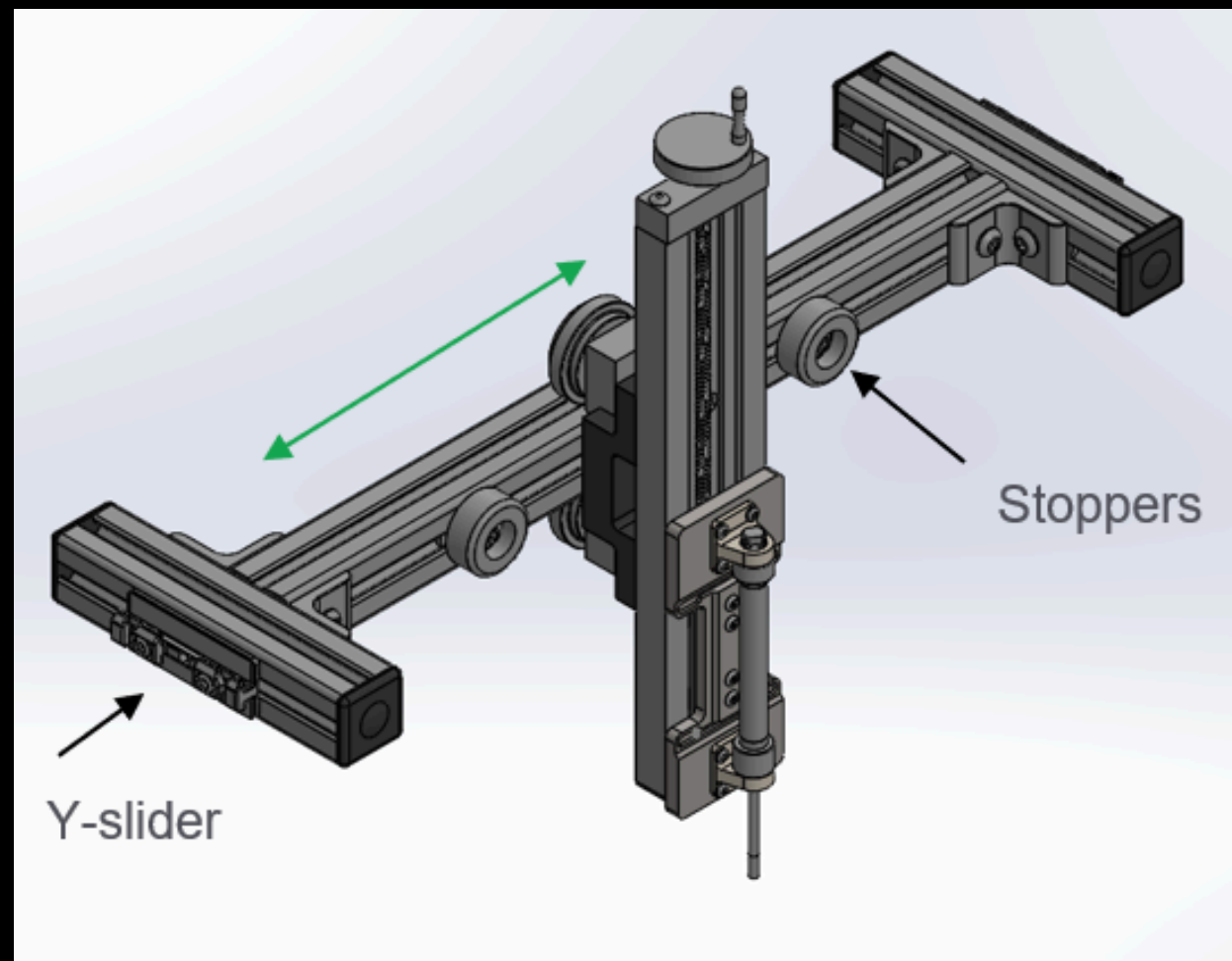
Key Features

- Adjustable X-Y positioning
- Lead-screw Z adjustment for precise press depth
- Pneumatic actuation for automated cycle testing
- Modular 80/20 aluminum frame
- Adjustable workpiece mounting
- Supports single- and multi- button testing



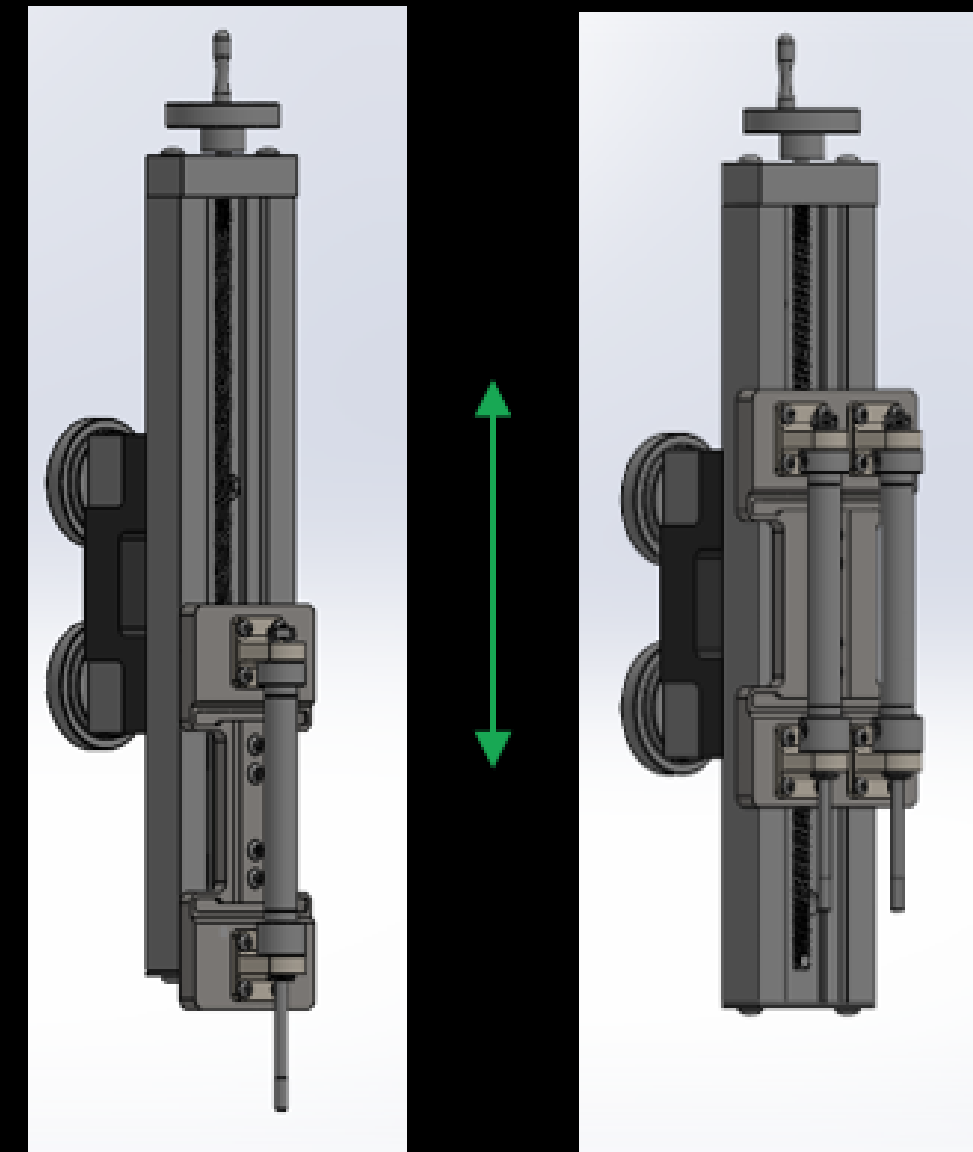
Button Cycle Tester

01



X-Y Positioning

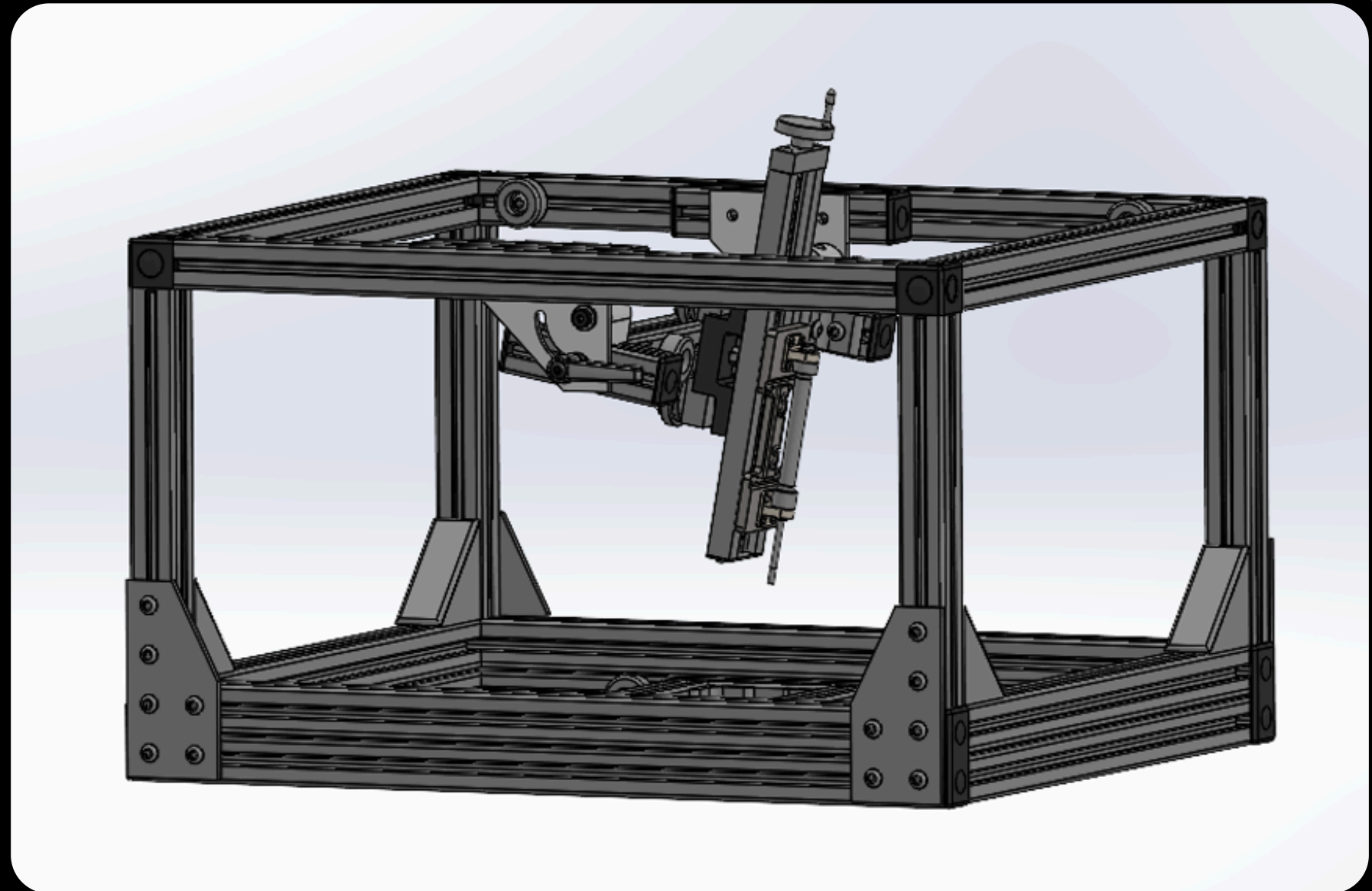
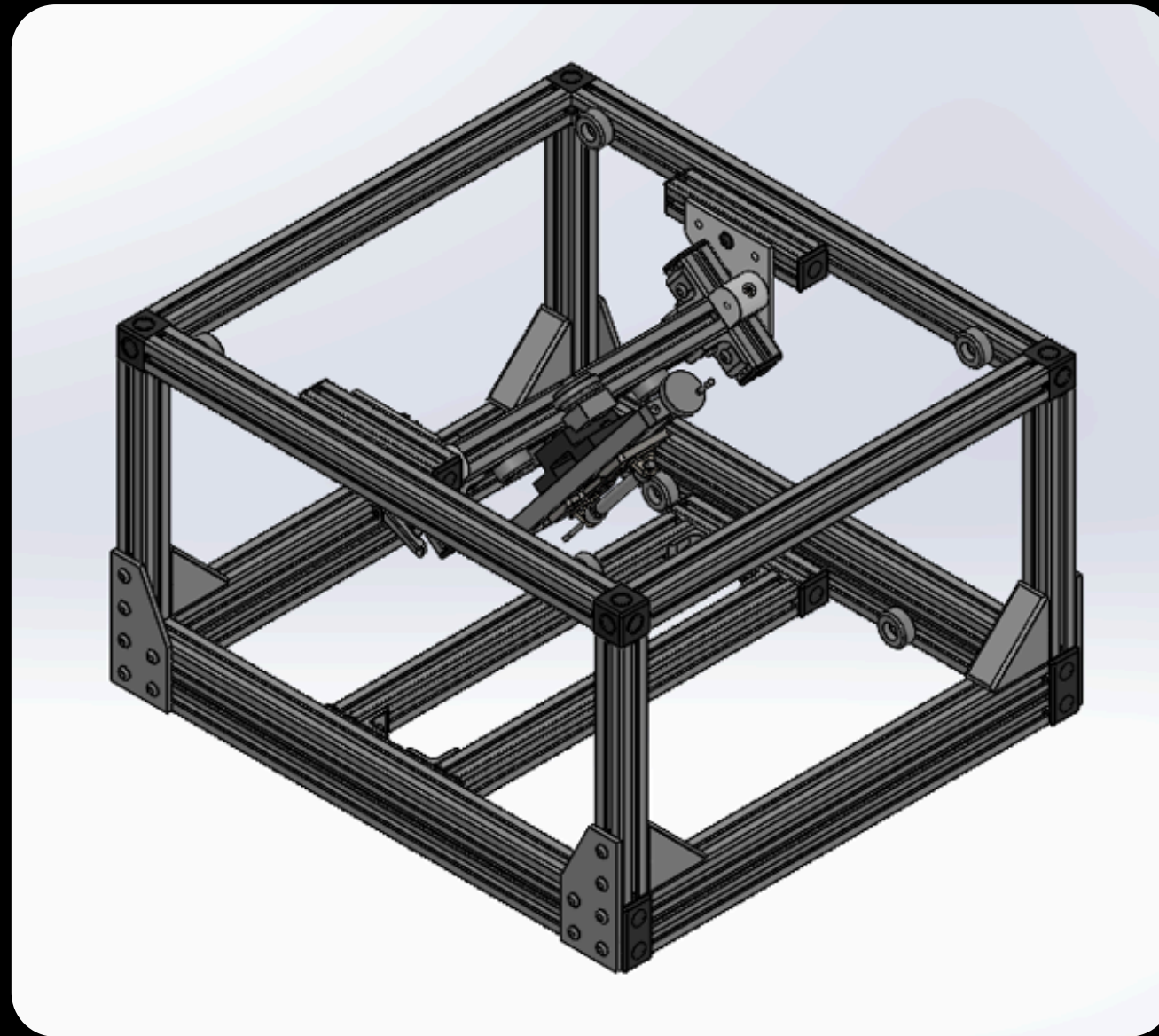
- Wide positioning range
- Repeatable mechanical stops



Single/dual-button configurations

Button Cycle Tester

01

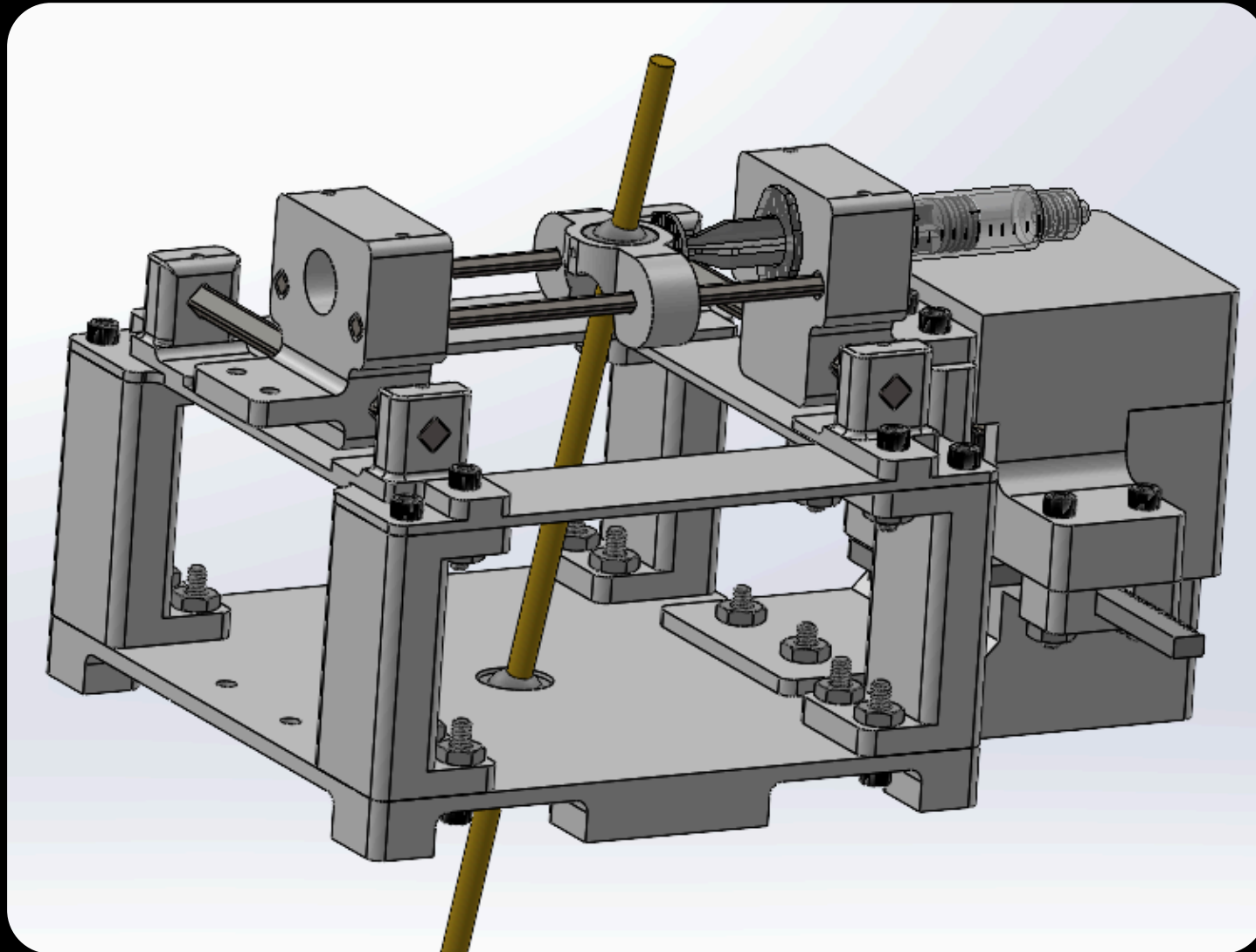


Tilter Add-On

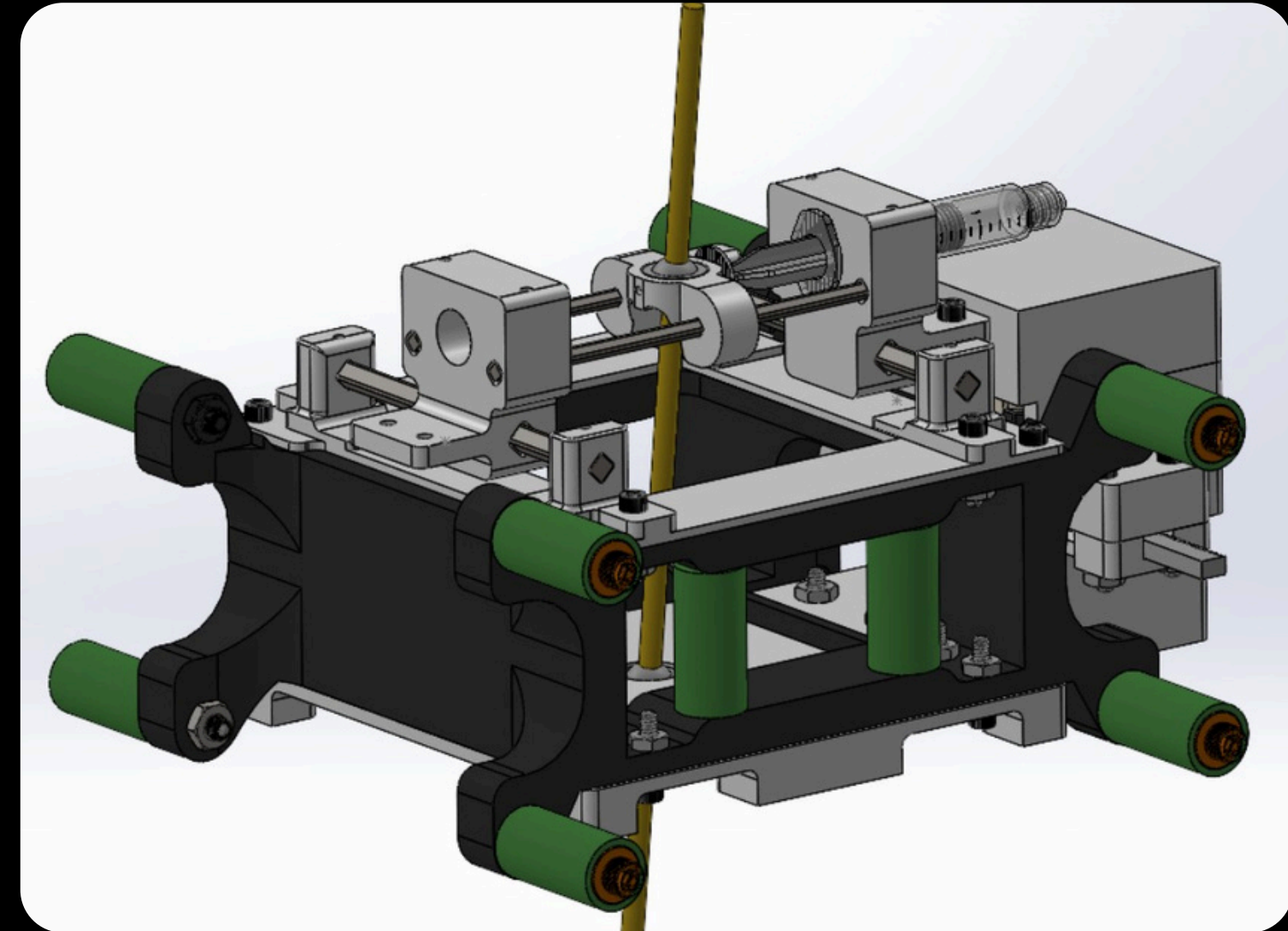
- Adjustable actuation angle
- Angled button alignment

MRI-Compatible Multi-Procedural Robot

02



End-Effector Assembly



Carriage Integration

Overview

Designed and prototyped the end-effector for an MRI-compatible multi-procedural robotic platform for image-guided needle- and probe-based procedures. The end-effector enables controlled positioning and angulation of surgical tools while integrating with the larger robotic system.

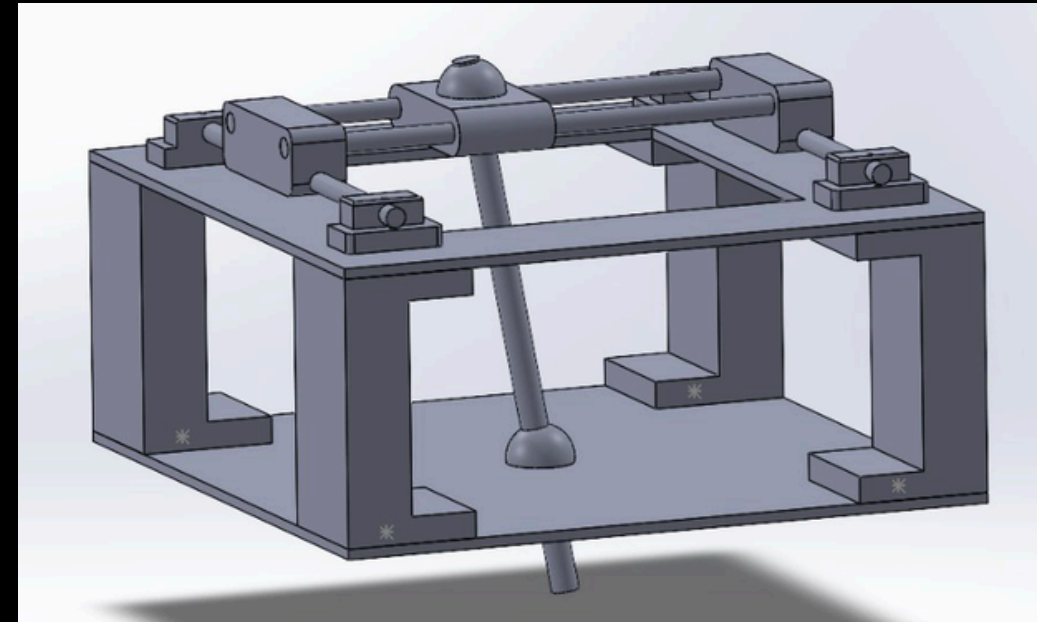
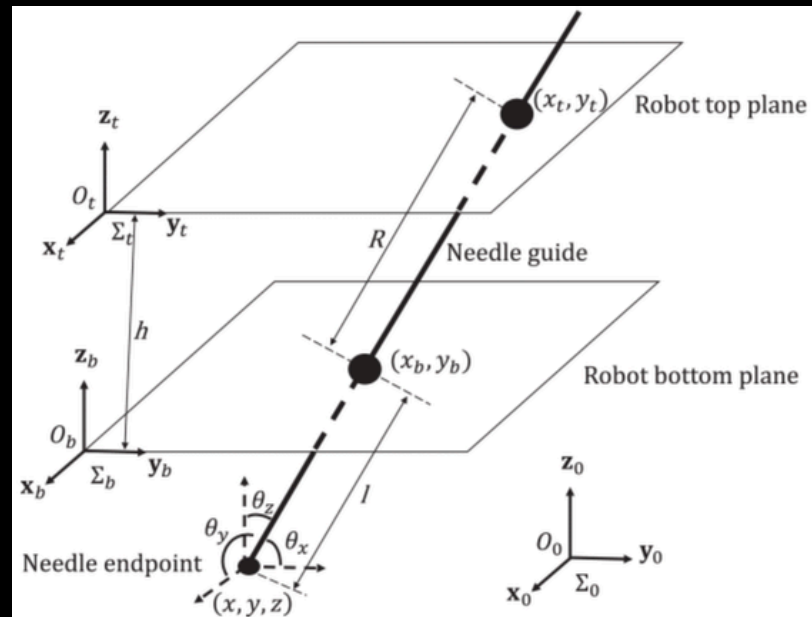
Key Features

- 2-DOF needle/probe positioning
- Up to 30° angular motion
- Rail-guided linear motion
- Modular integration with robot carriage



MRI-Compatible Multi-Procedural Robot

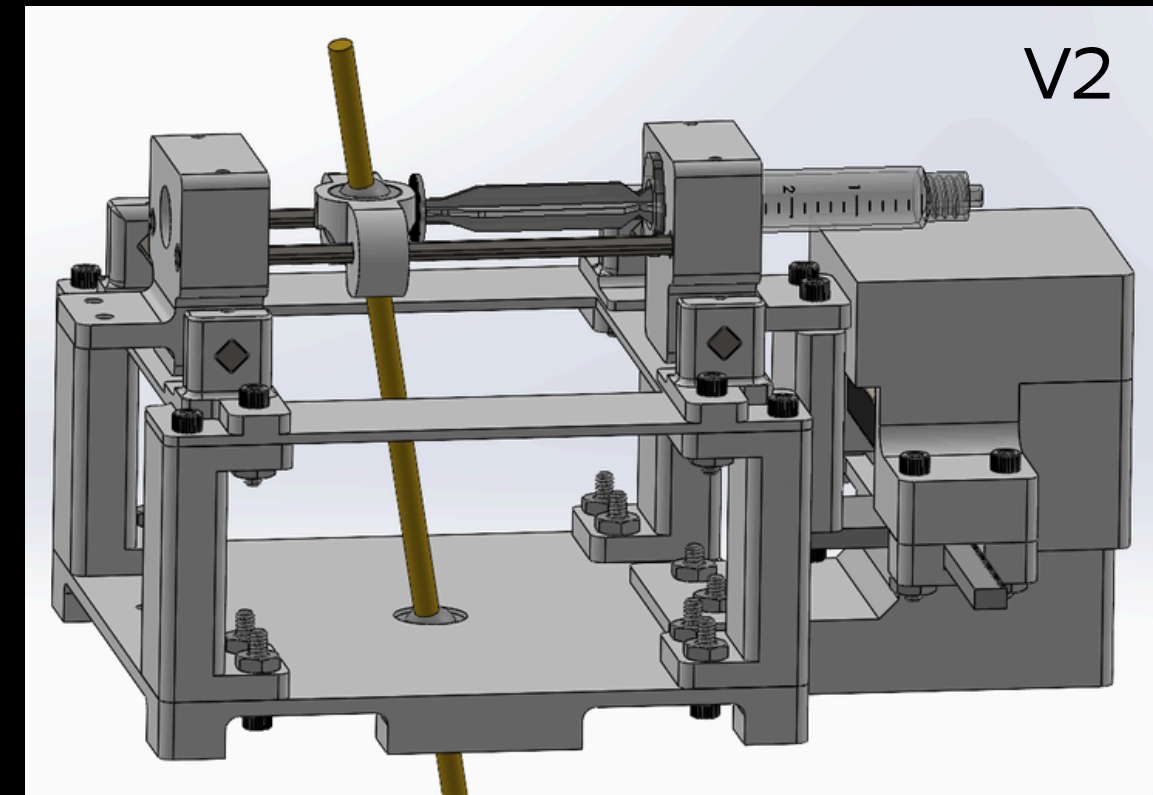
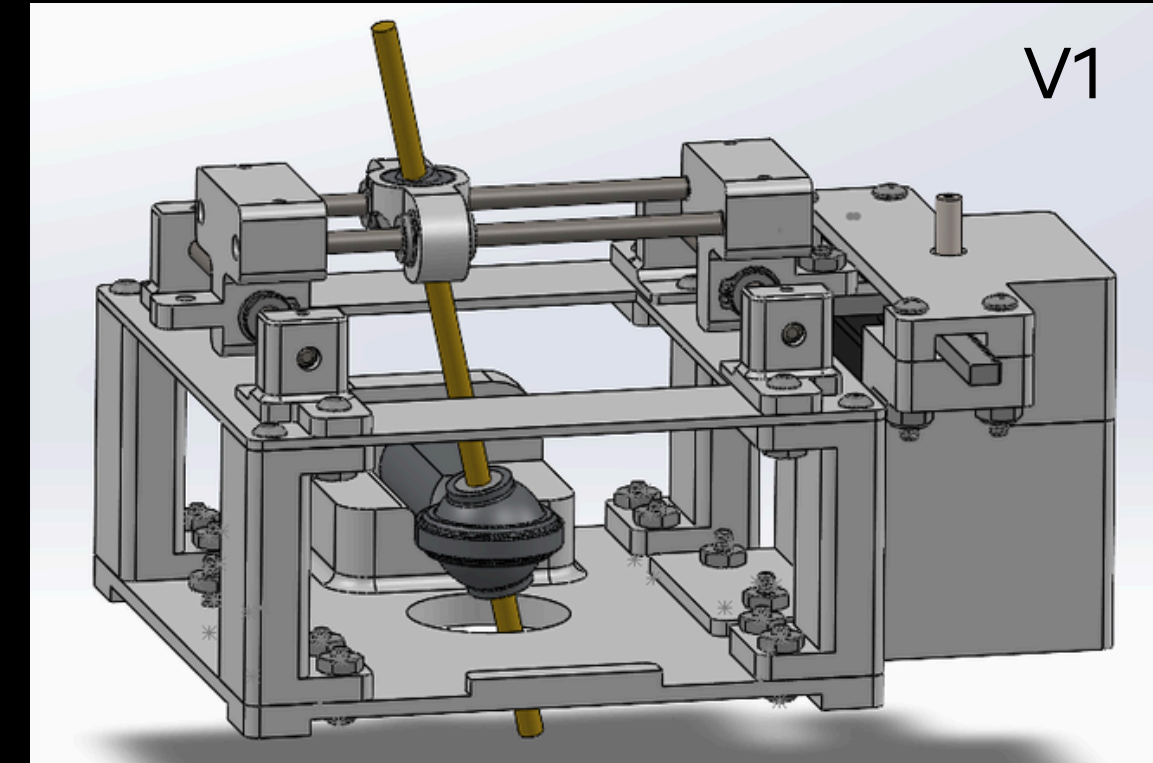
02



Concept Design

Design Iteration

Developed the initial 2-DOF end-effector concept into manufacturable designs through iterative CAD. Refined the rail system, test actuator mounting, structural geometry, and assembly layout to improve motion, alignment, and integration with the robot carriage.

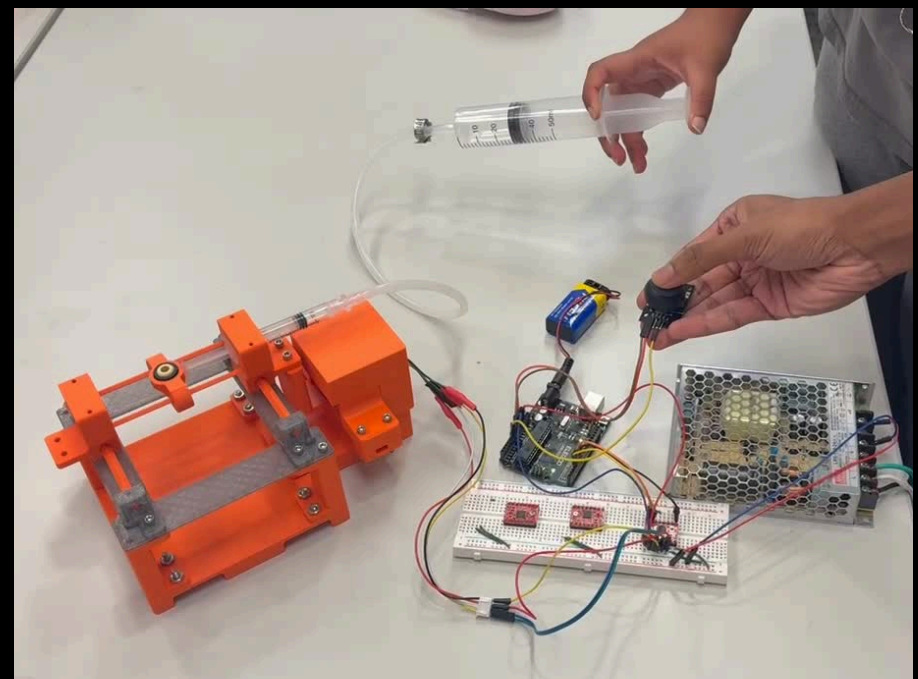
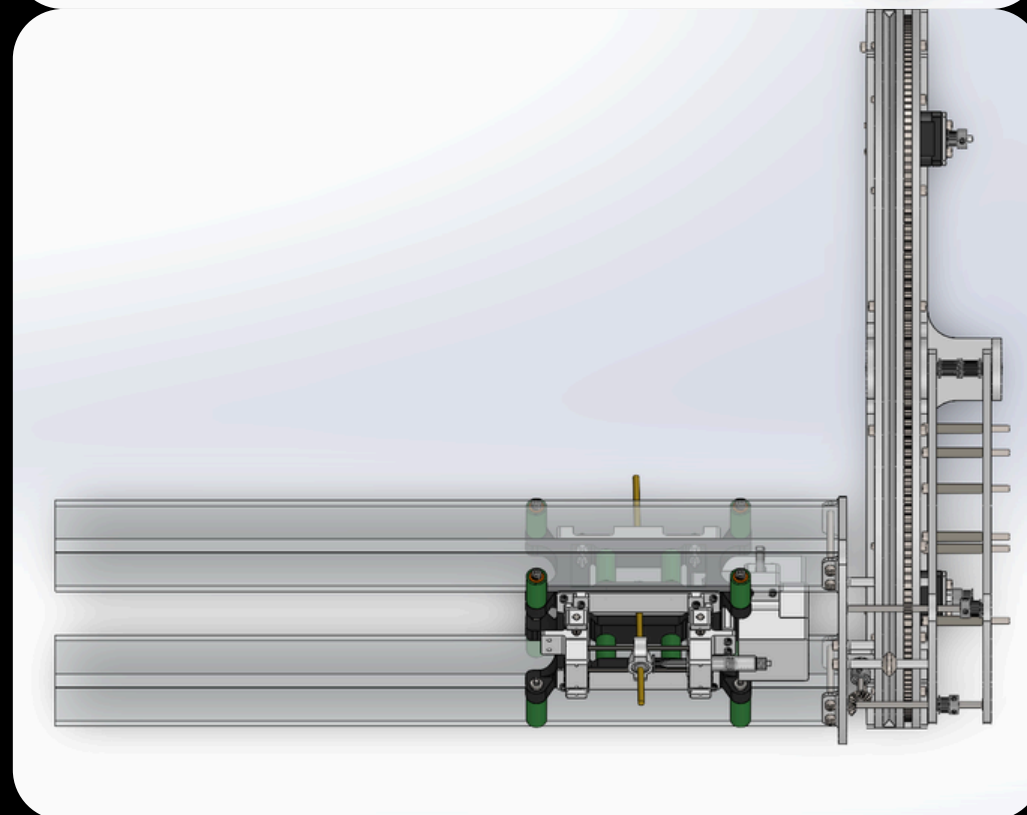
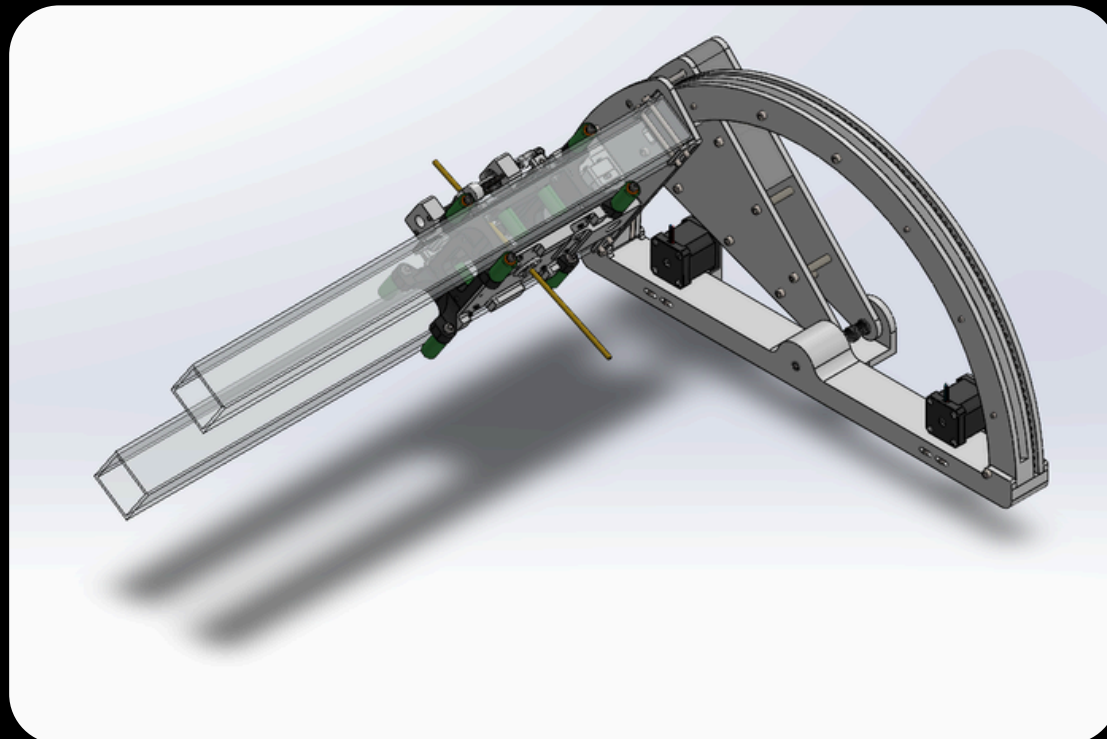
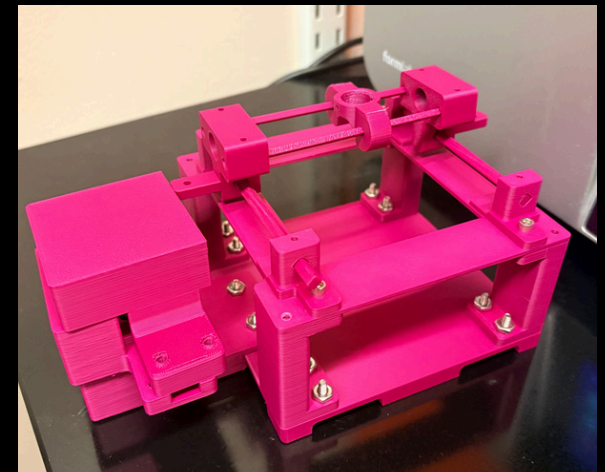
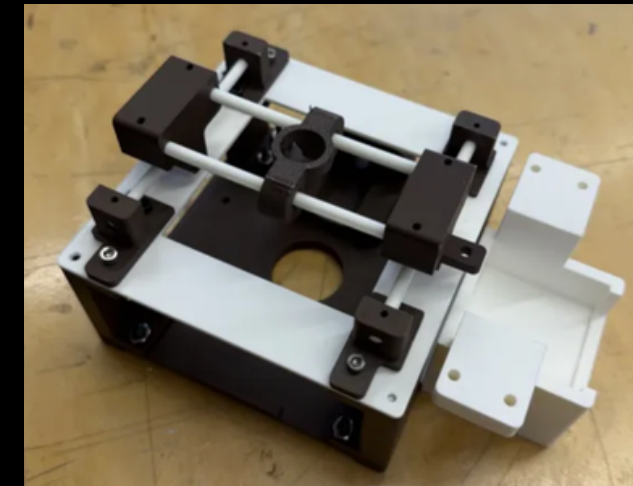
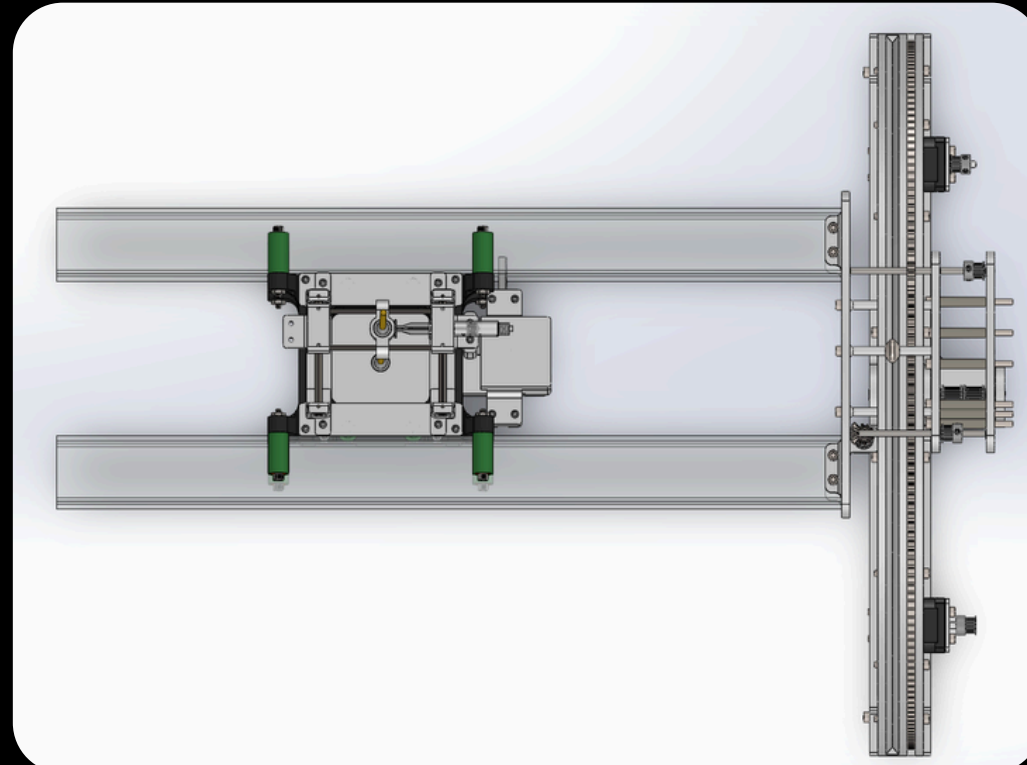
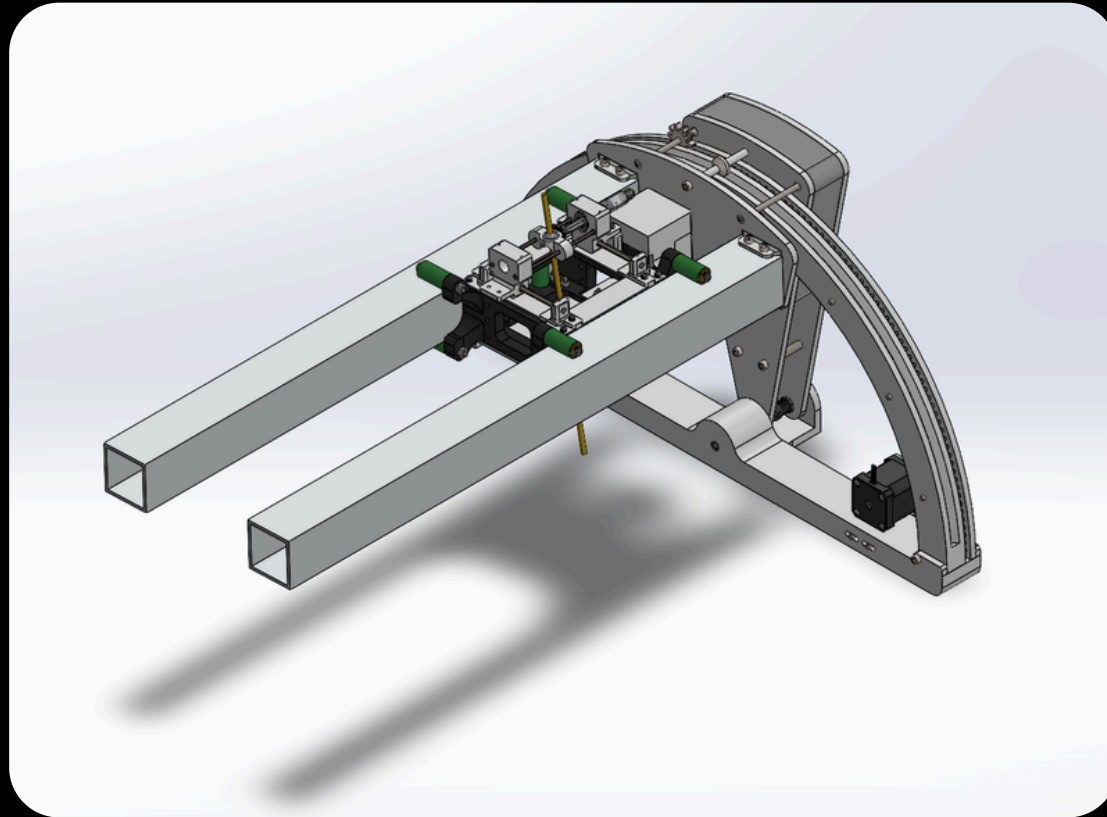


Manufacturable Design Iterations



MRI-Compatible Multi-Procedural Robot

02



System Integration

Iterative Prototyping



Short-Circuit Test Fixture

03

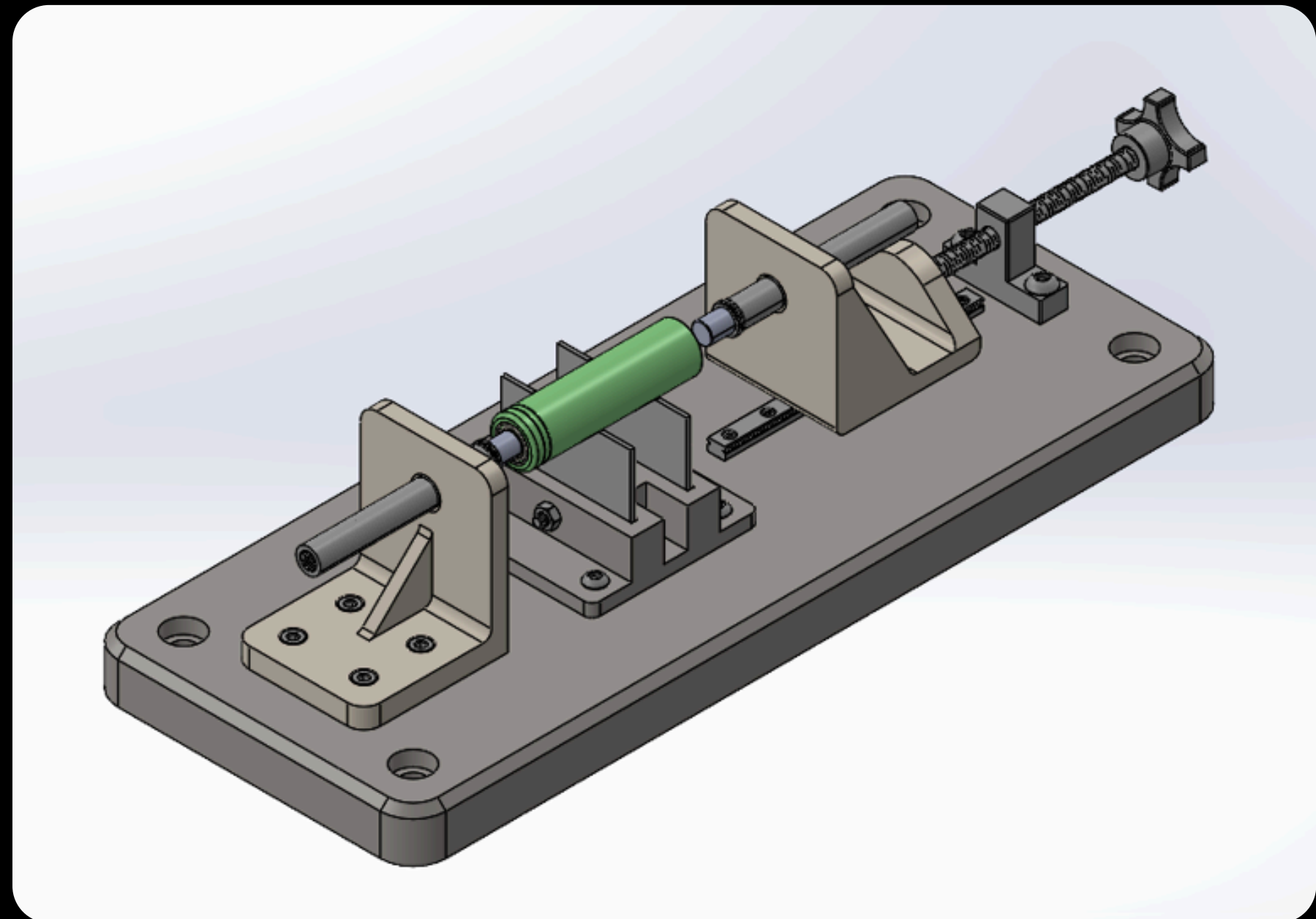
Overview

Designed a fixture to perform controlled short-circuit testing of cylindrical battery cells used across MSA products, supporting intrinsic safety testing and evaluation of cell behavior.

Developed as a repeatable, modular test platform supporting multiple standard cell sizes with controlled probe engagement.

Key Features

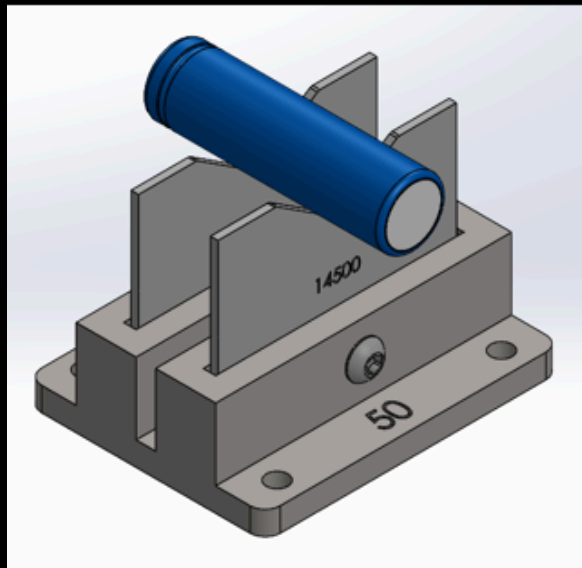
- 14500, 18650, 21700, 26650, 32650 cell compatibility
- Spring-loaded Gorilla probes for consistent contact
- Screw-driven linear probe adjustment
- Modular test setup



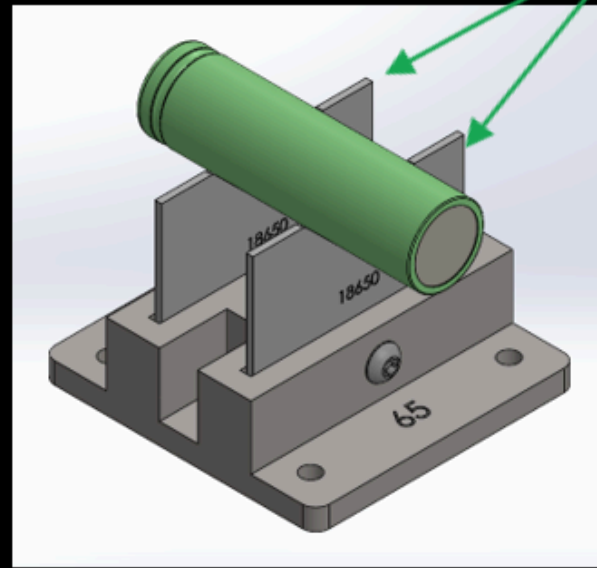
Short-Circuit Test Fixture

03

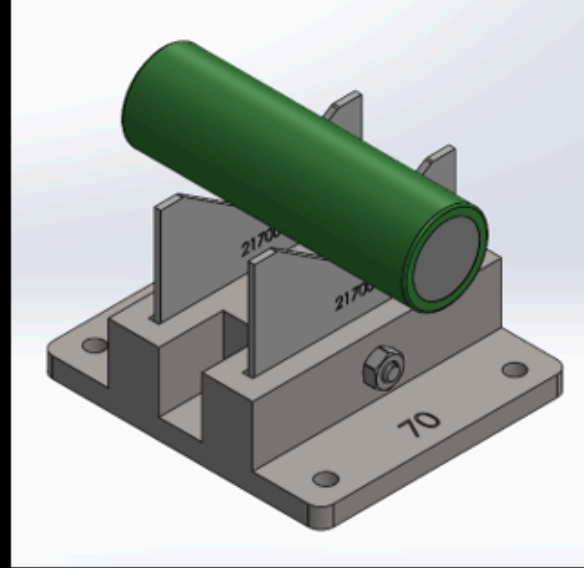
FR4 Cell Holders



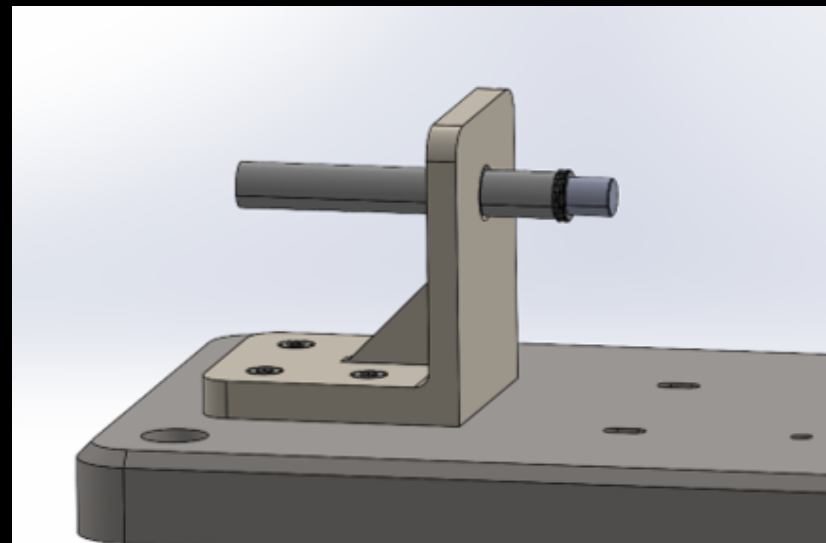
14500 Cell



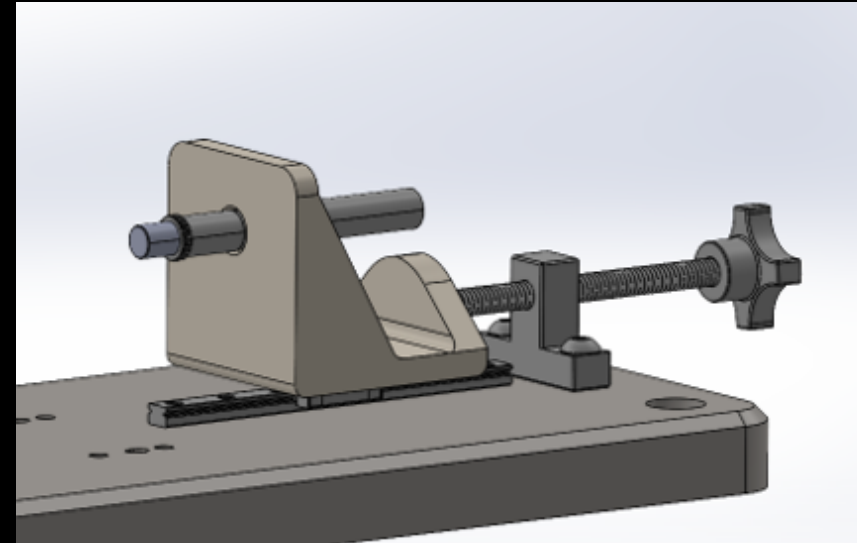
18650 Cell



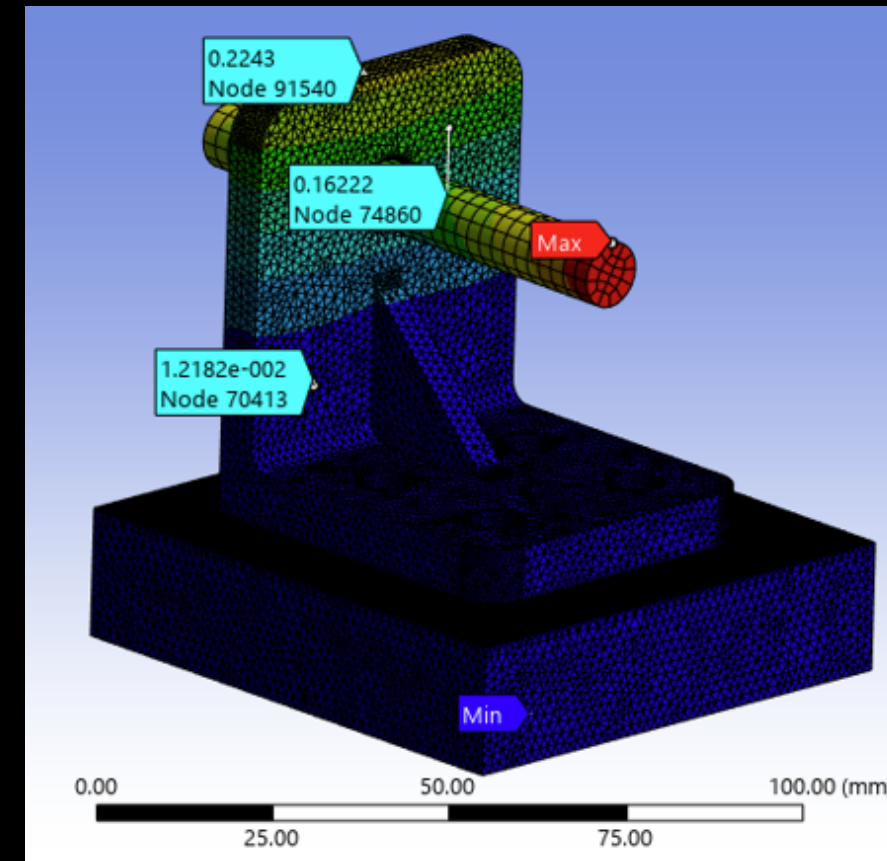
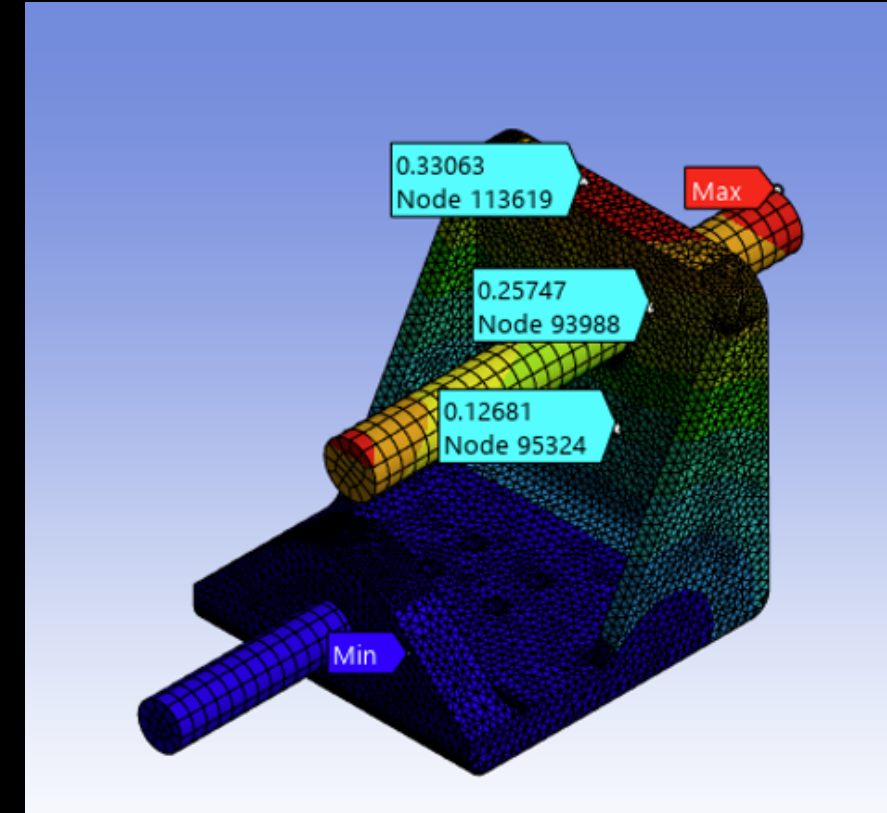
21700 Cell



Stationary Probe



Screw-Driven Probe

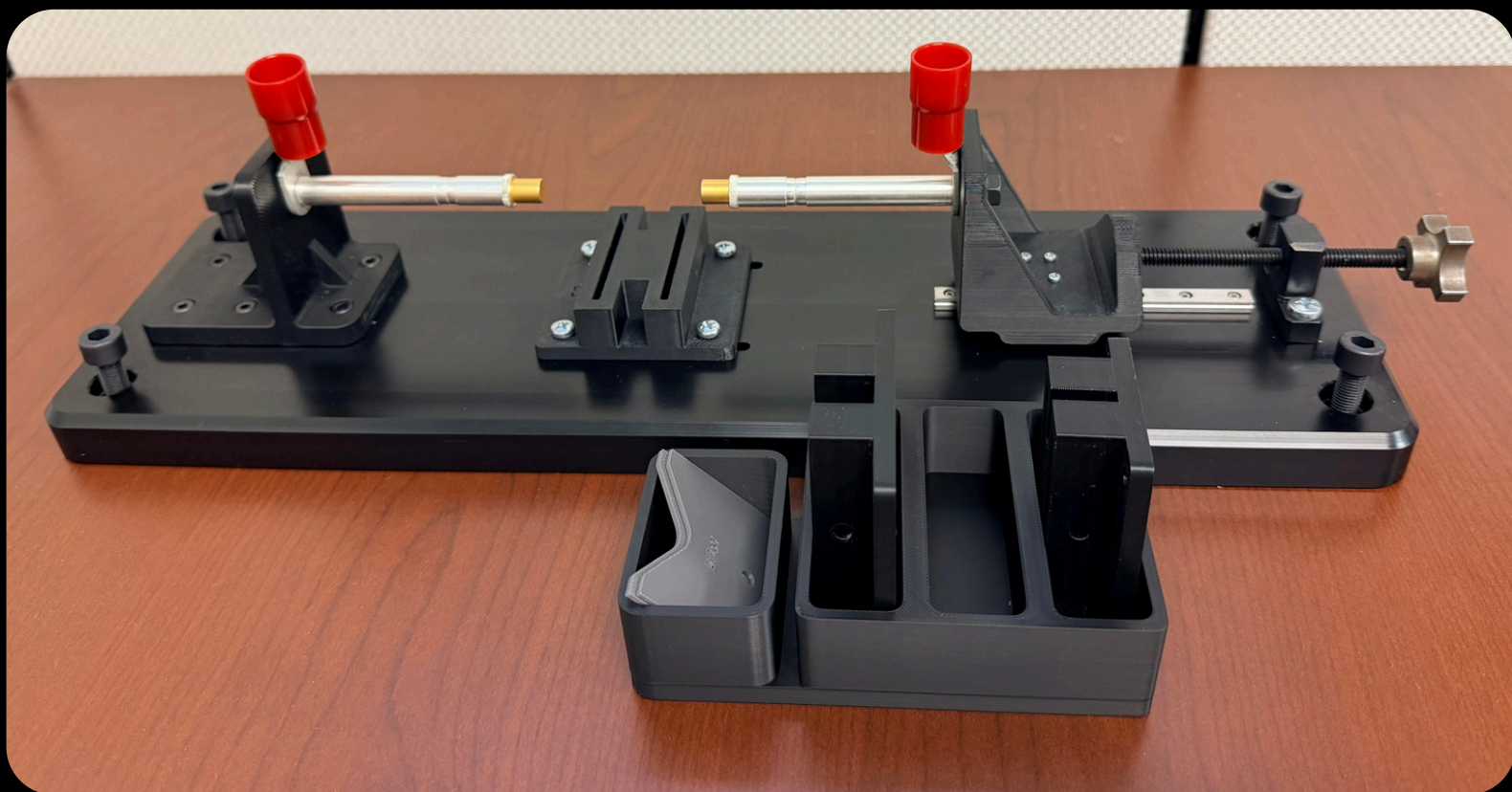
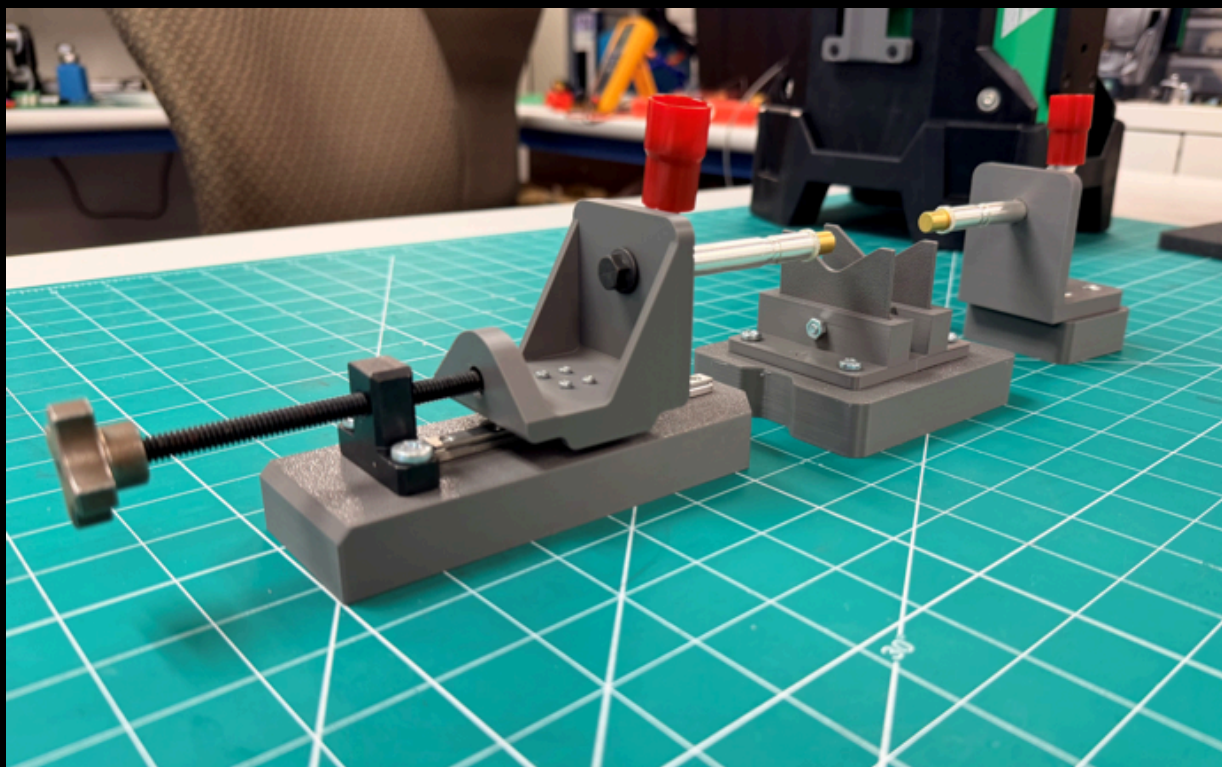
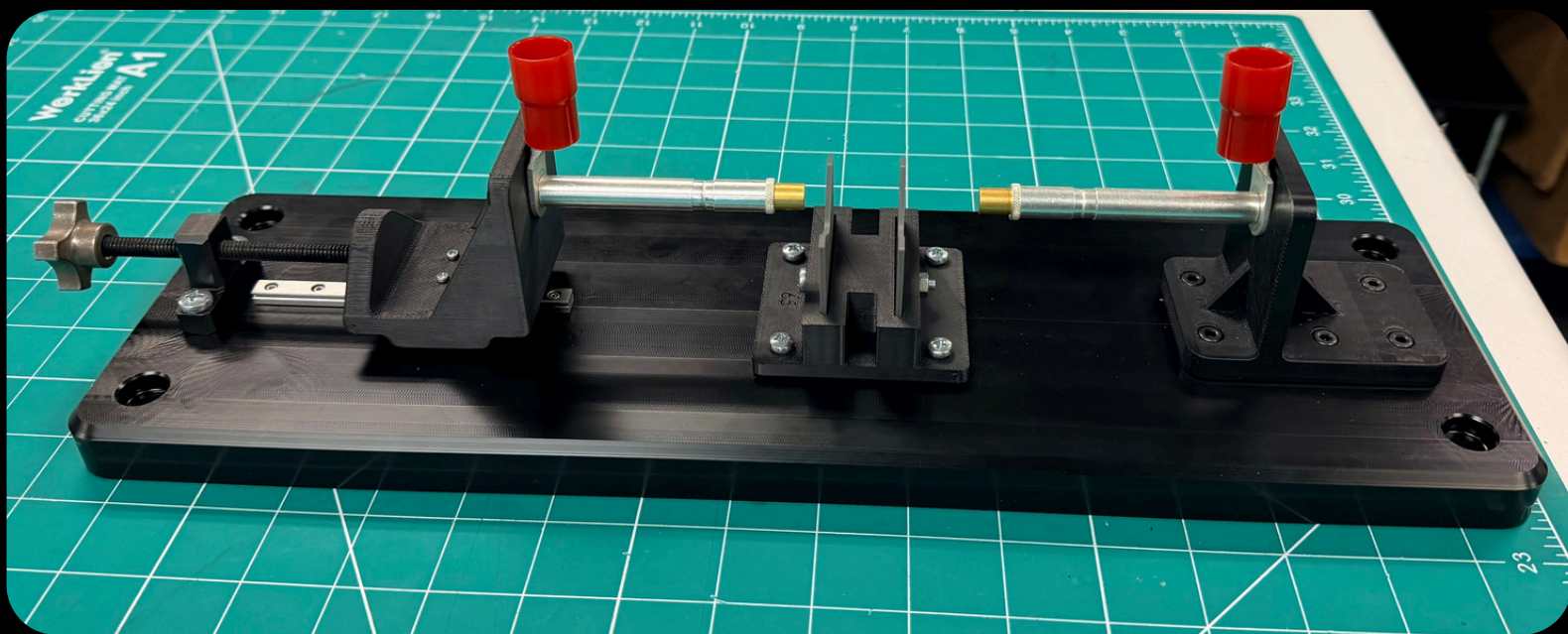
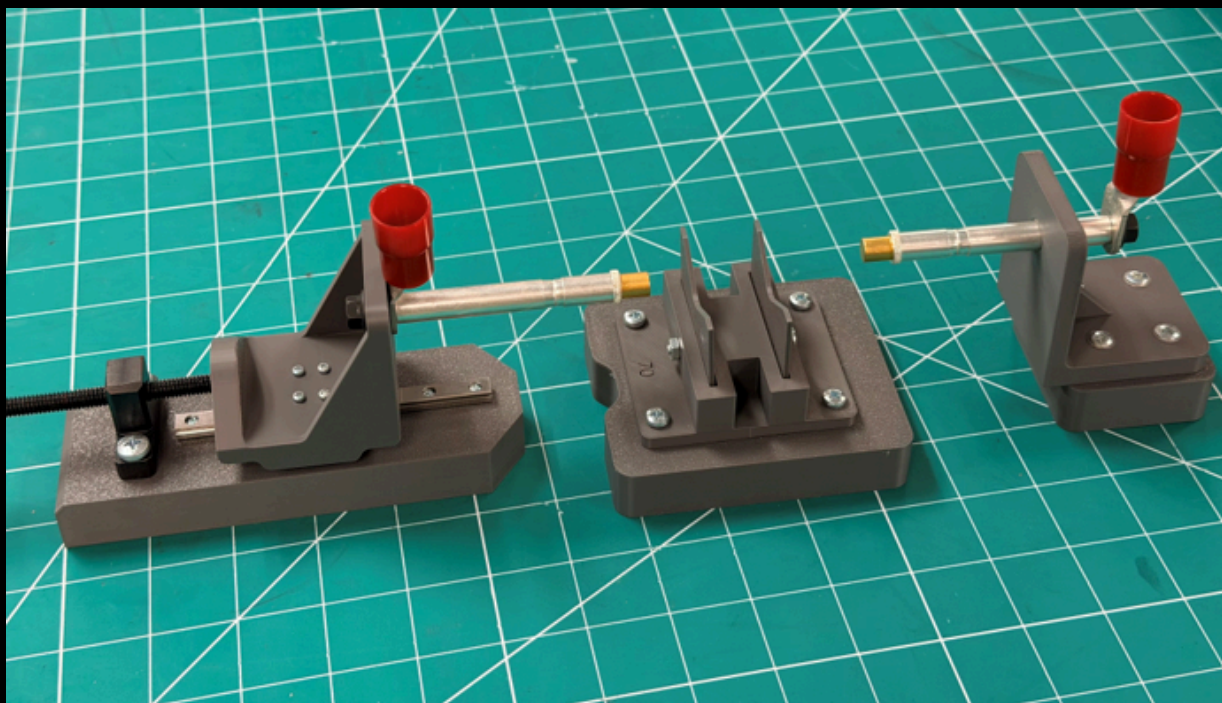


ANSYS FEA

Used ANSYS Mechanical to evaluate bracket deflection under probe loading and optimize rib geometry for increased stiffness. Iteratively prototyped brackets to refine stiffness, tolerances, and alignment

Short-Circuit Test Fixture

03



Prototype

Final Fixture

Sterile Post-Processing Station

04

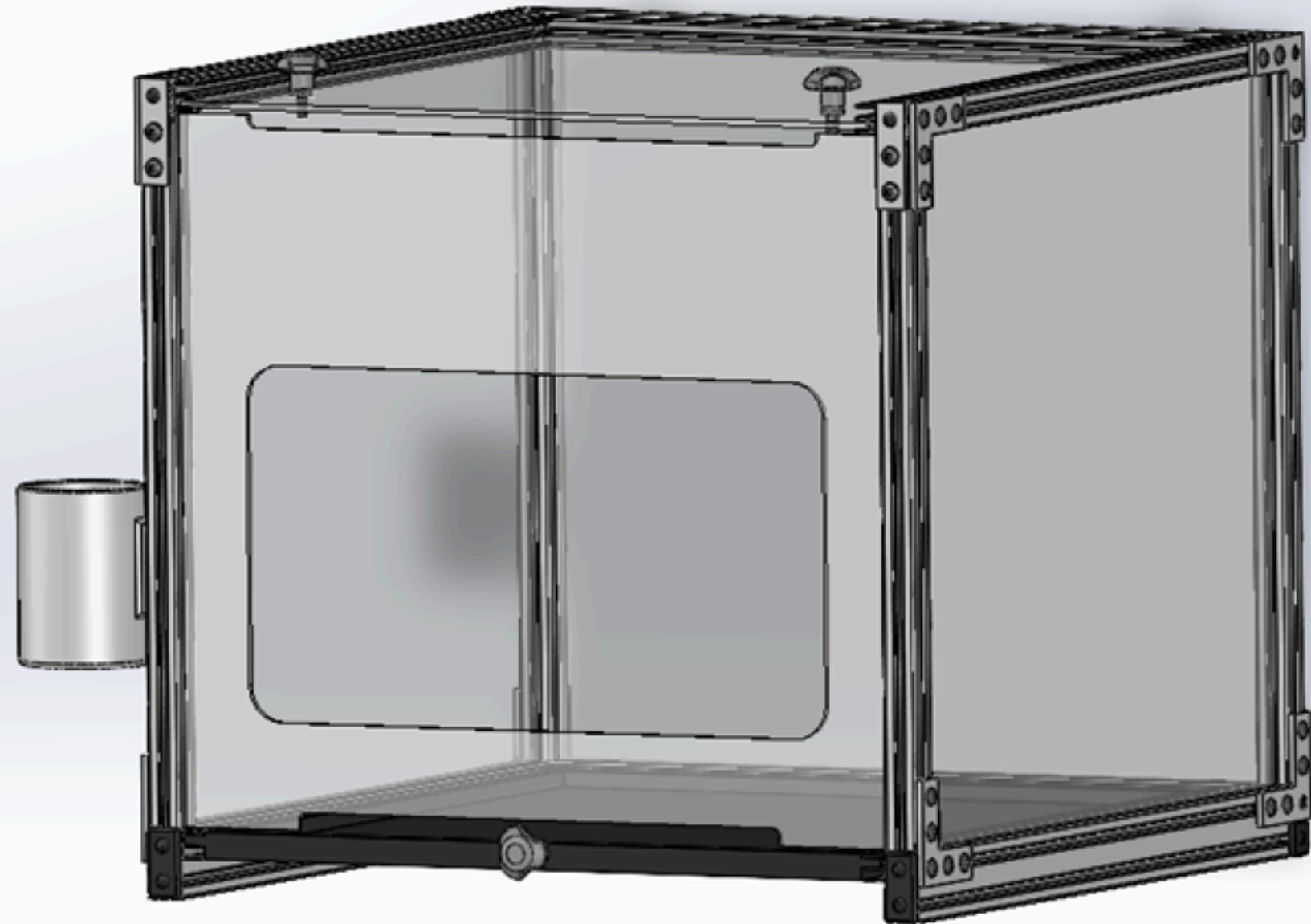
Overview

Designed a cleanroom-compatible workstation to safely post-process sterile surgical guide resin models in an ISO-7 cleanroom.

Developed to improve workflow efficiency and operator safety while providing a dedicated workspace for sterile resin model post-processing.

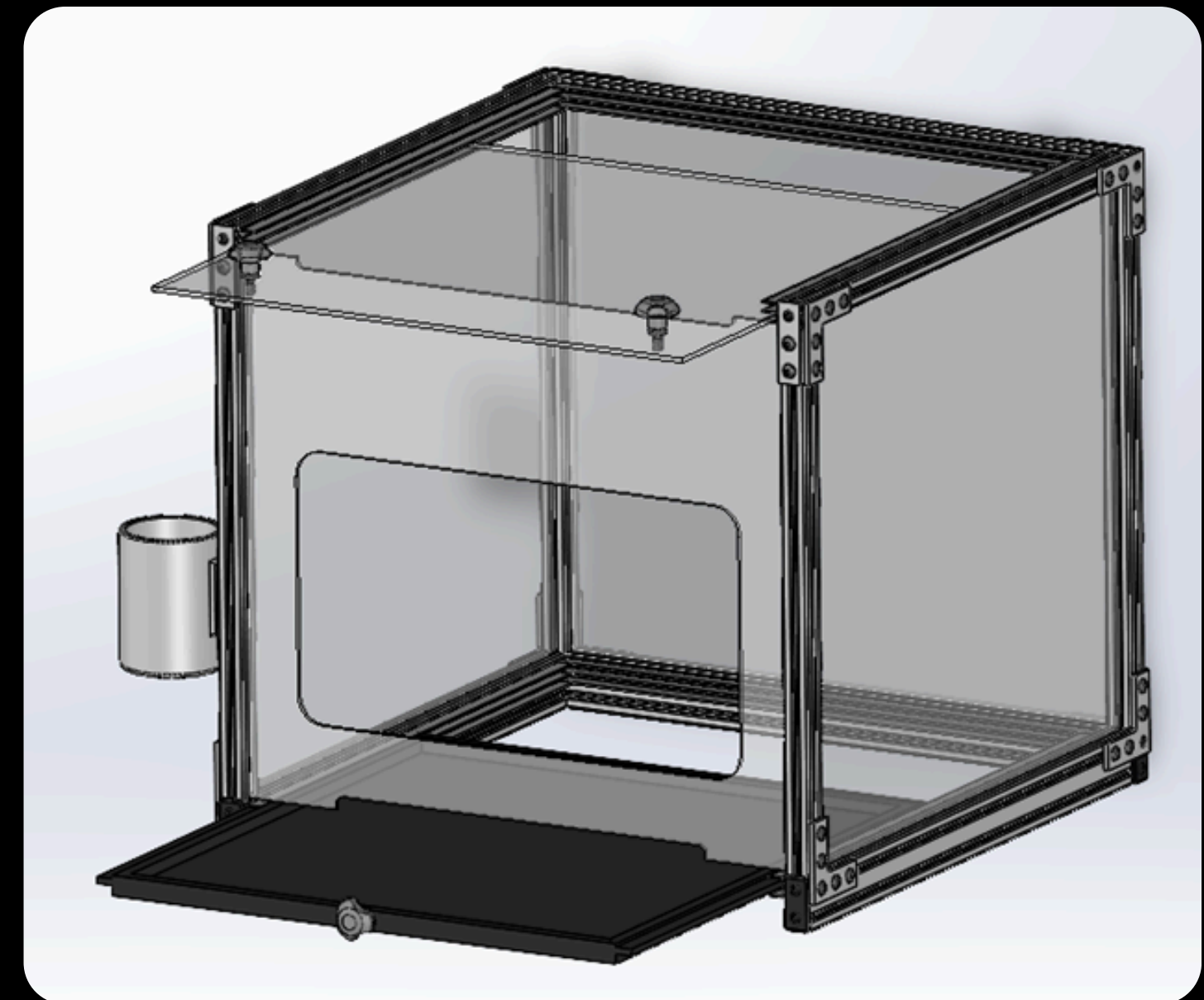
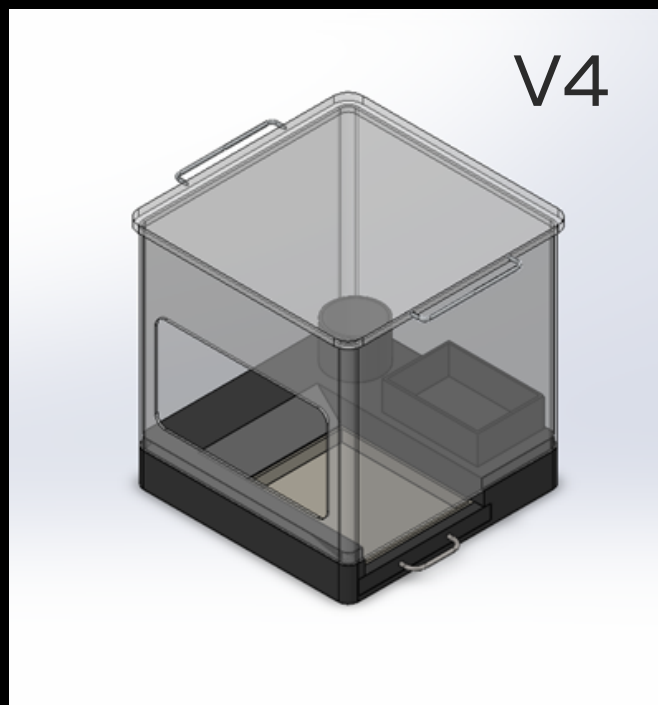
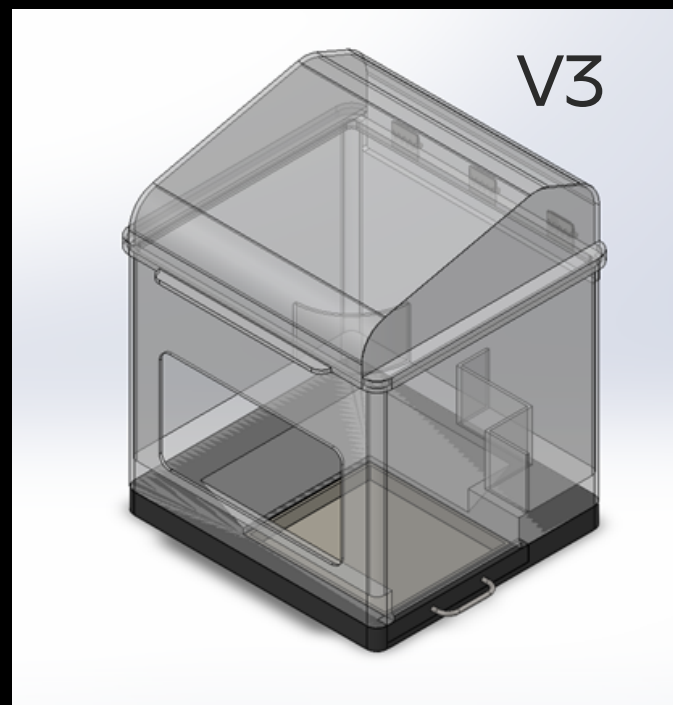
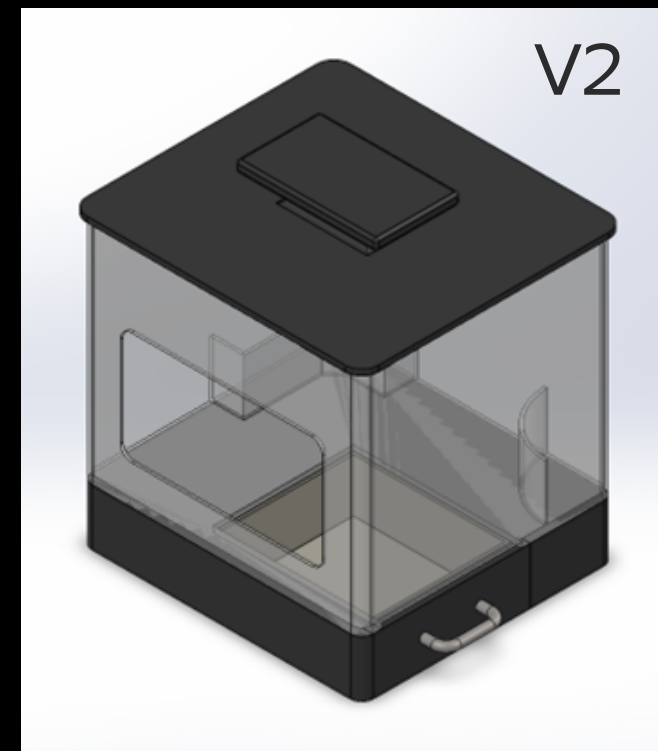
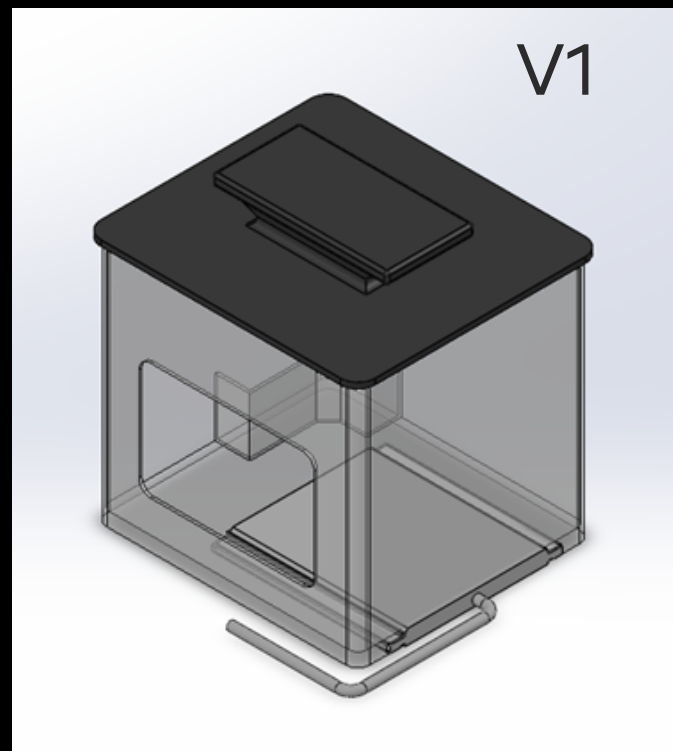
Key Features

- ISO-7 cleanroom-compatible design
- 80/20 smooth aluminum extrusion frame
- IPA-resistant PVC-Glas panels
- Enclosed post-processing workspace
- Integrated waste disposal
- Front access sized for operator reach



Sterile Post-Processing Station

04



Final Design

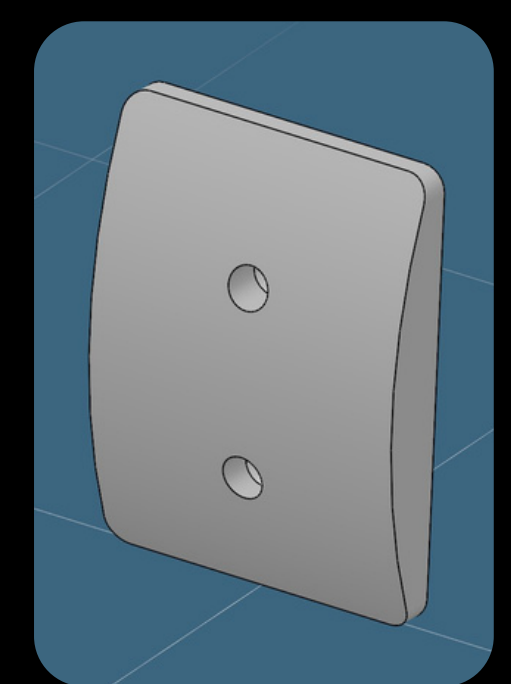
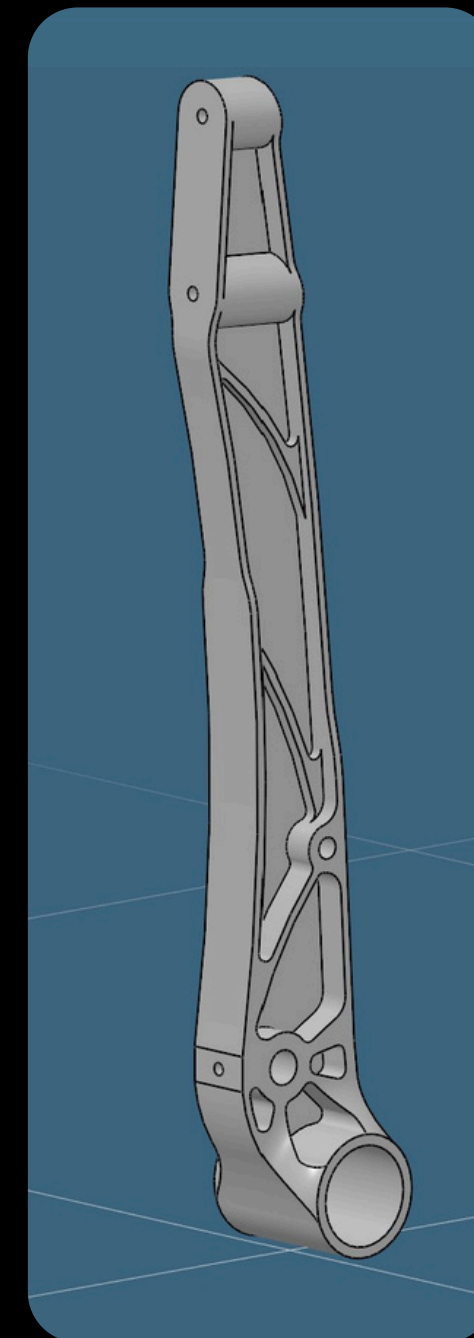
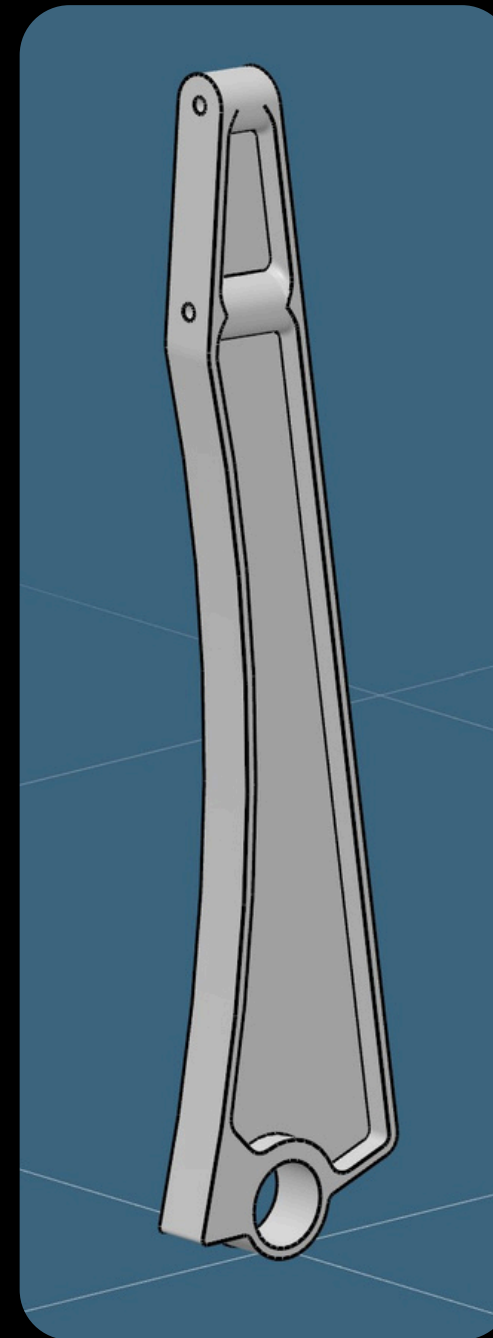
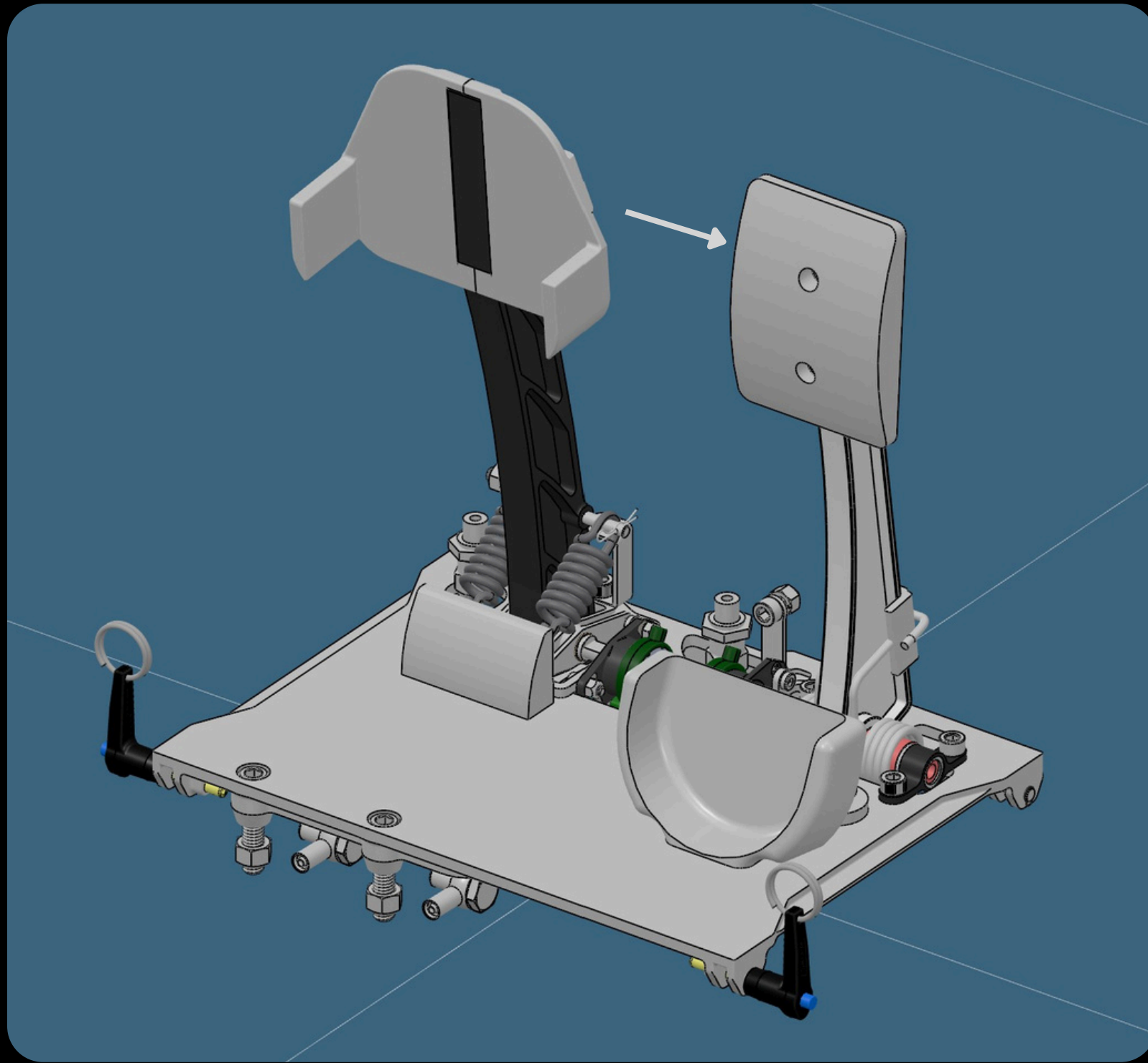
Explored 4+ concepts to refine waste disposal, feature placement, and workstation layout. The final design prioritized manufacturability, modularity, cost efficiency, and cleanroom compatibility.

Design Iterations

GT FSAE EV

HT09 Pedal Box Assembly

05

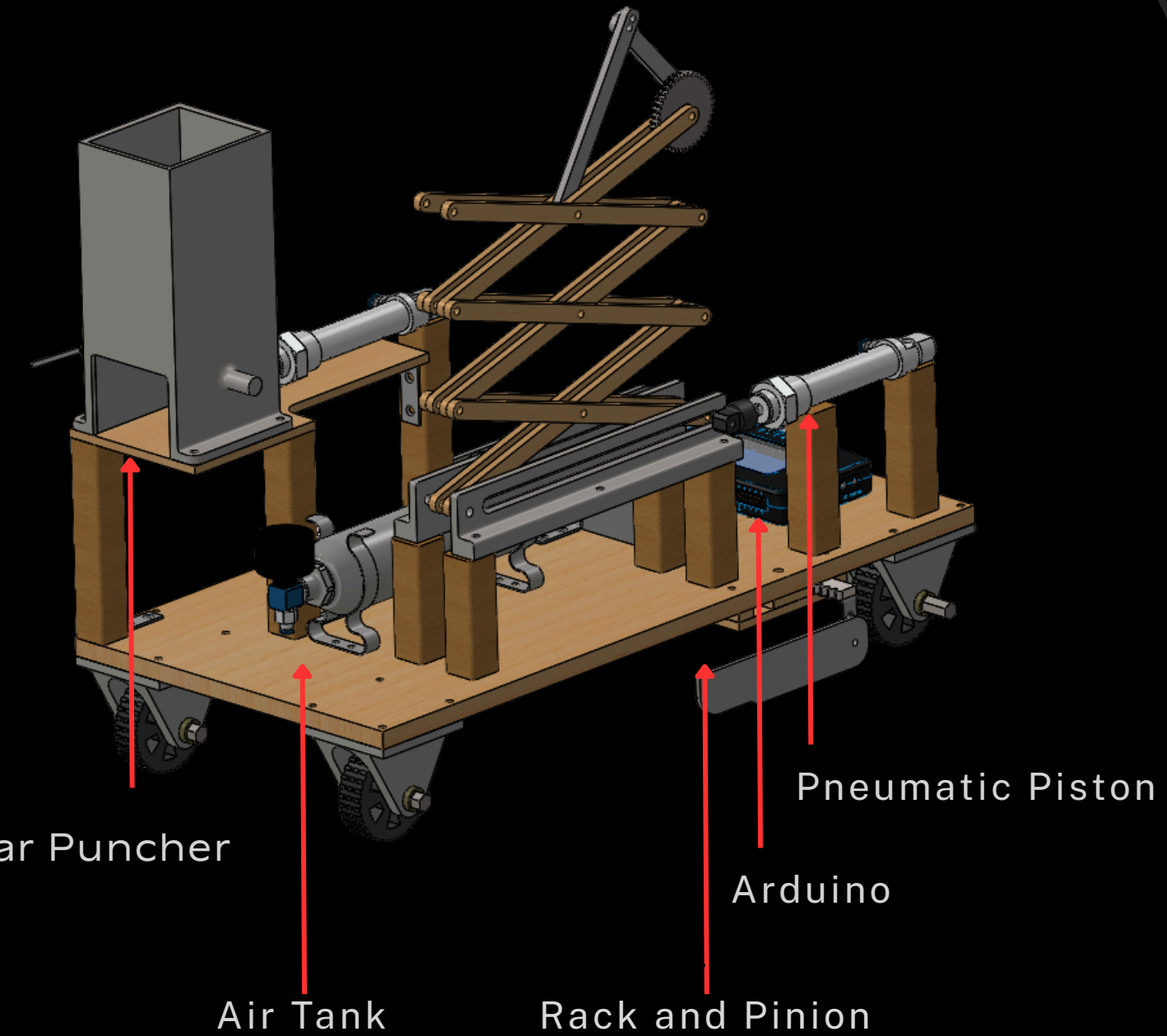
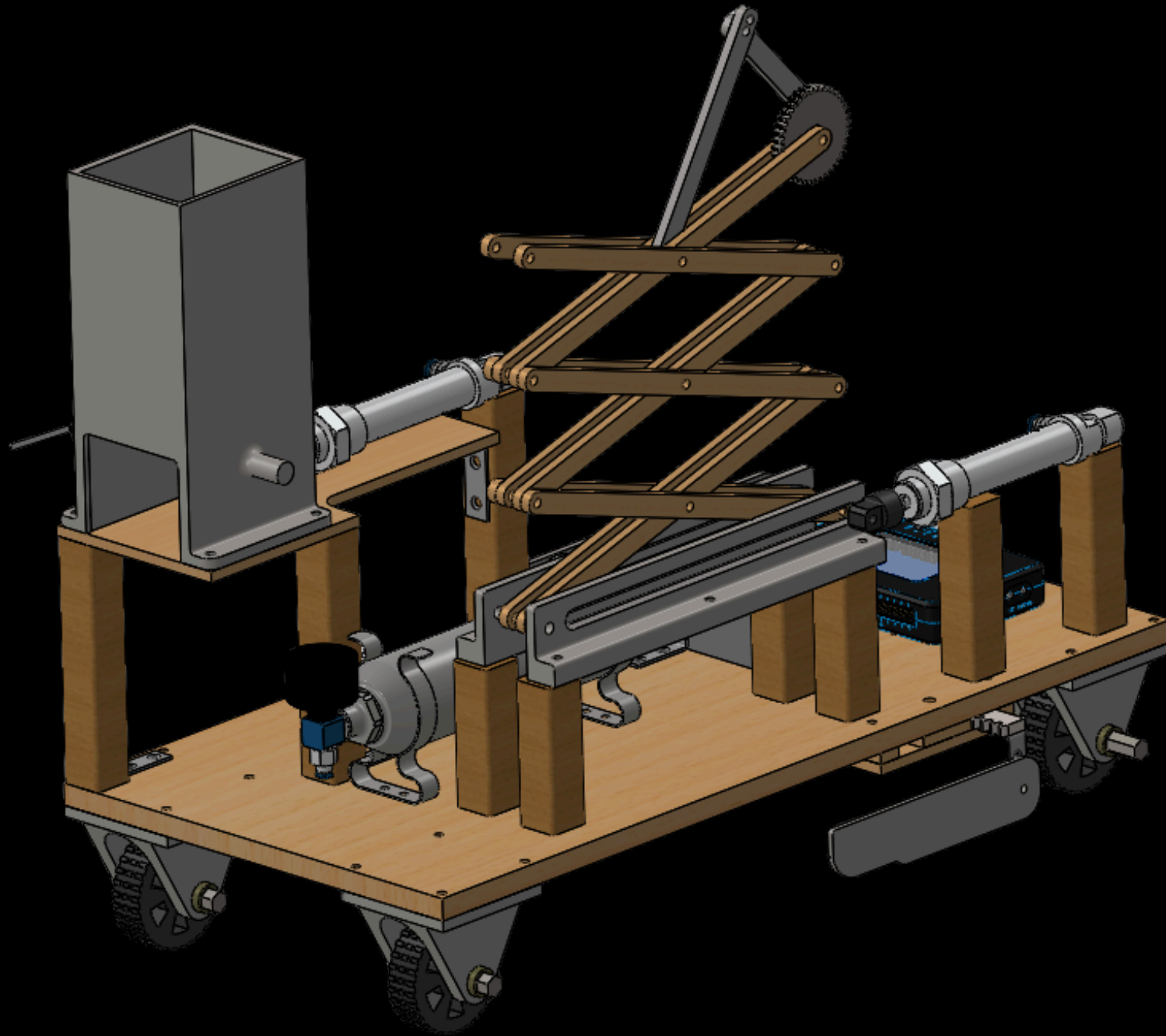


Redesigned the HT09 pedal box to simplify toe plate mounting and improve pedal performance. Replaced the previous two-piece side-mounted toe plate setup with a single-piece design, reducing part count and simplifying assembly. Redesigned the pedals around the new mounting scheme and optimized their geometry to reduce weight while improving driver ergonomics.



ME 2110 Autonomous Robot

06



Overview

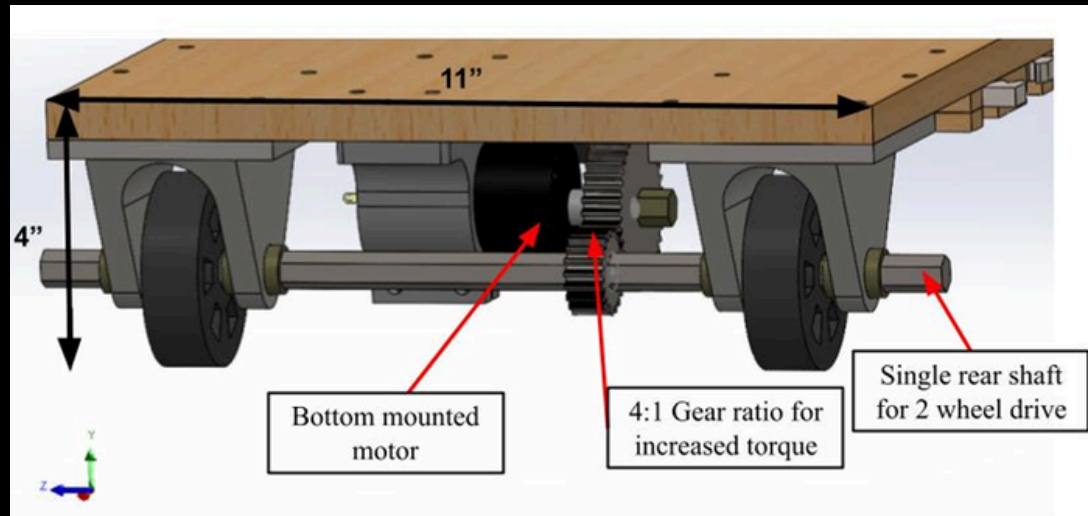
Designed and built an autonomous robot for the Georgia Tech ME 2110 design competition, where teams develop a robot to complete multiple scoring tasks within a timed course.

Our robot combined mechanical scoring mechanisms, pneumatics, sensors, and Arduino-based controls to autonomously navigate the course and execute multi-step scoring sequences.

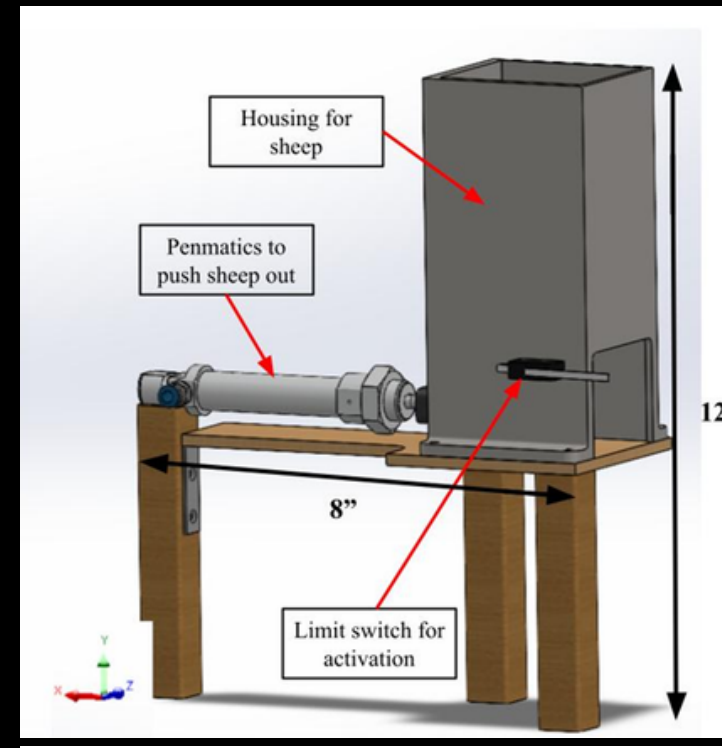


ME 2110 Autonomous Robot

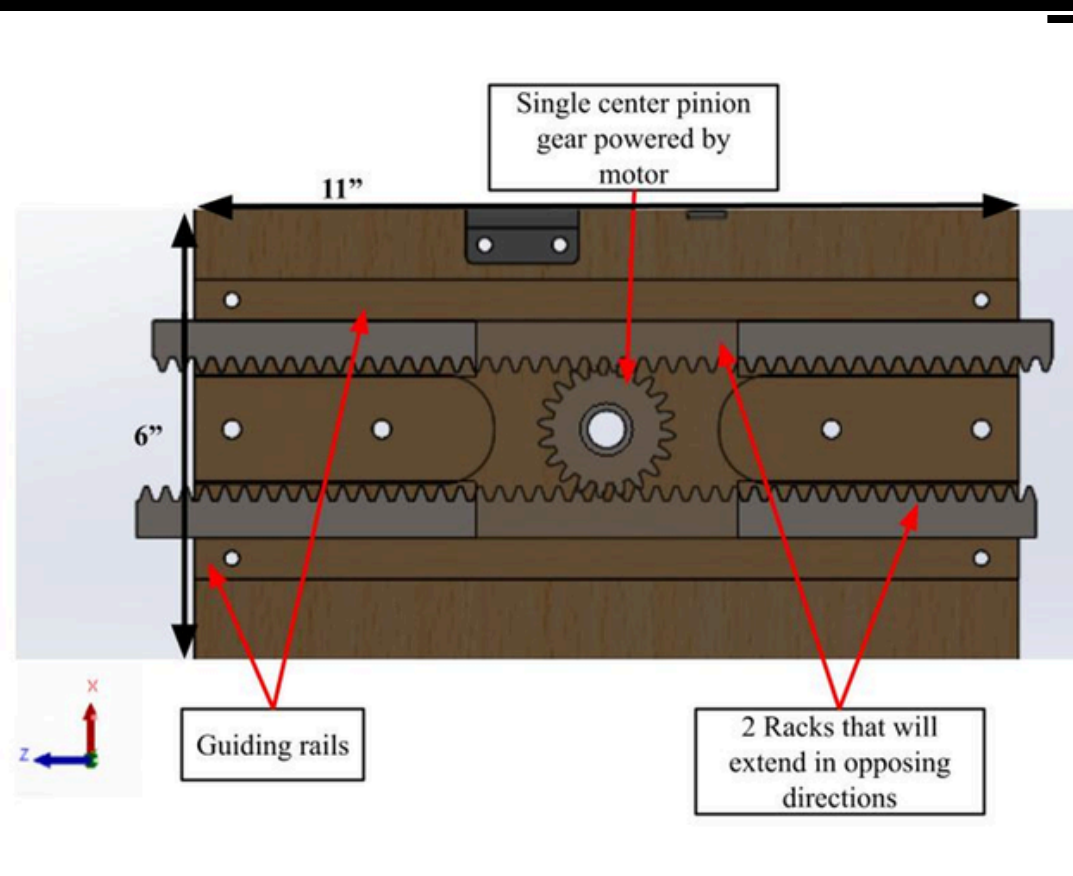
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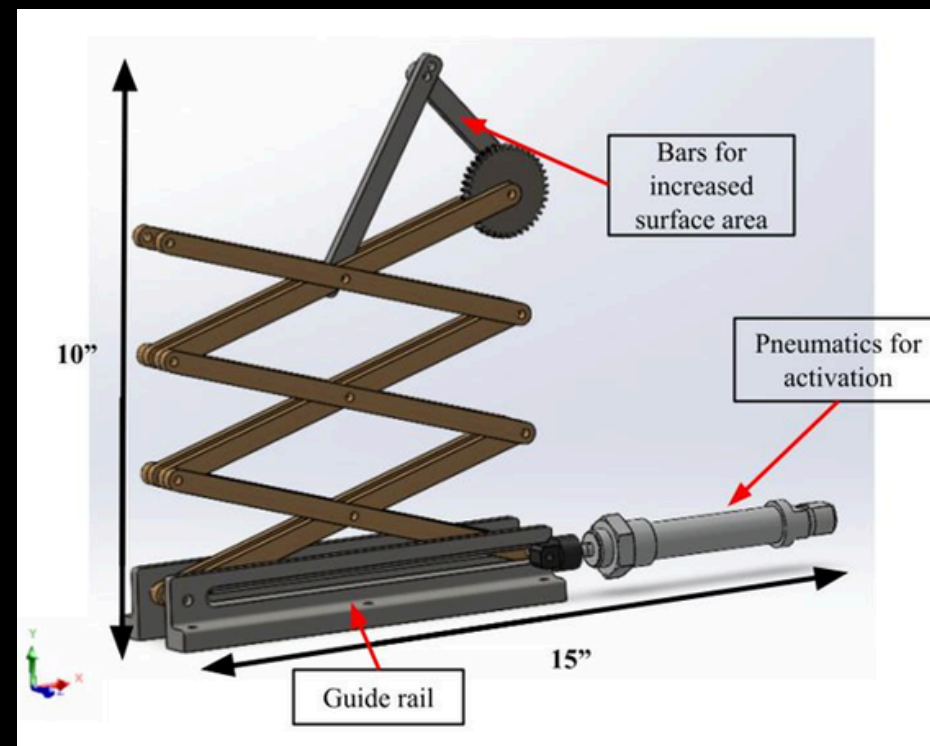
Drivetrain



Linear Puncher



Rack & Pinion



Scissor Lift

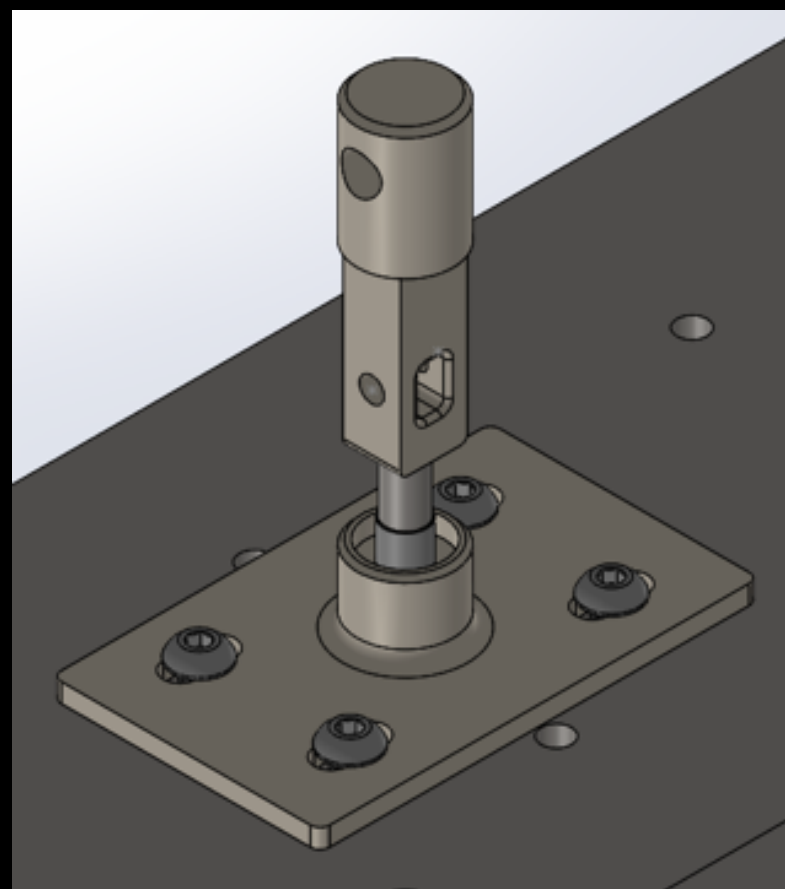
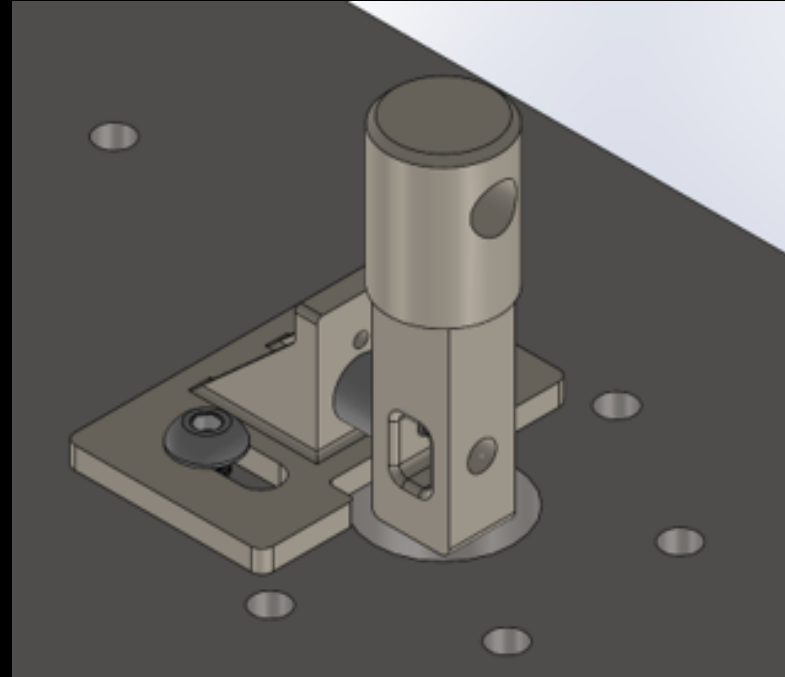


Final Robot

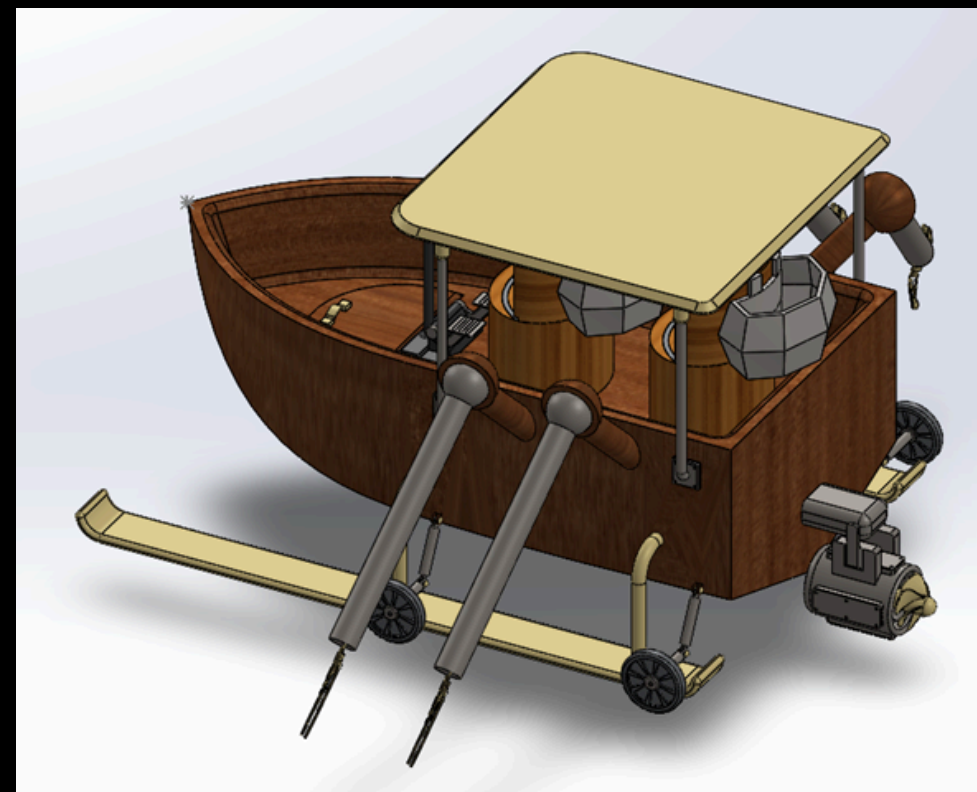
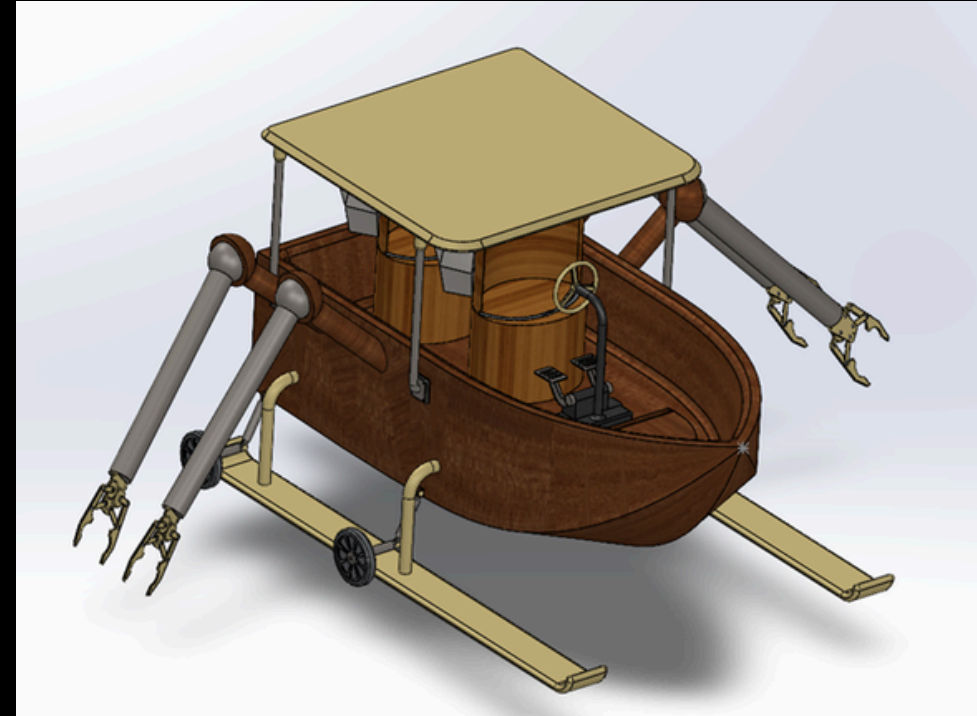
Ranked 5th/72 pre-competition • 95.9% scoring reliability

Additional CAD Projects

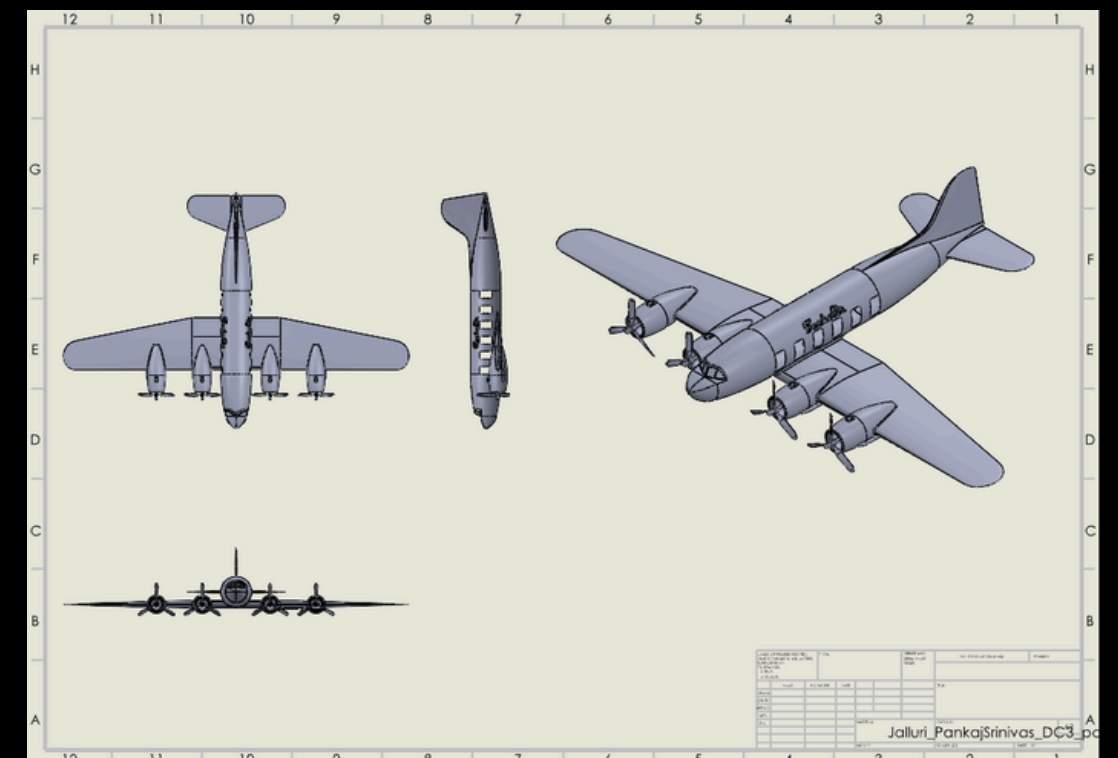
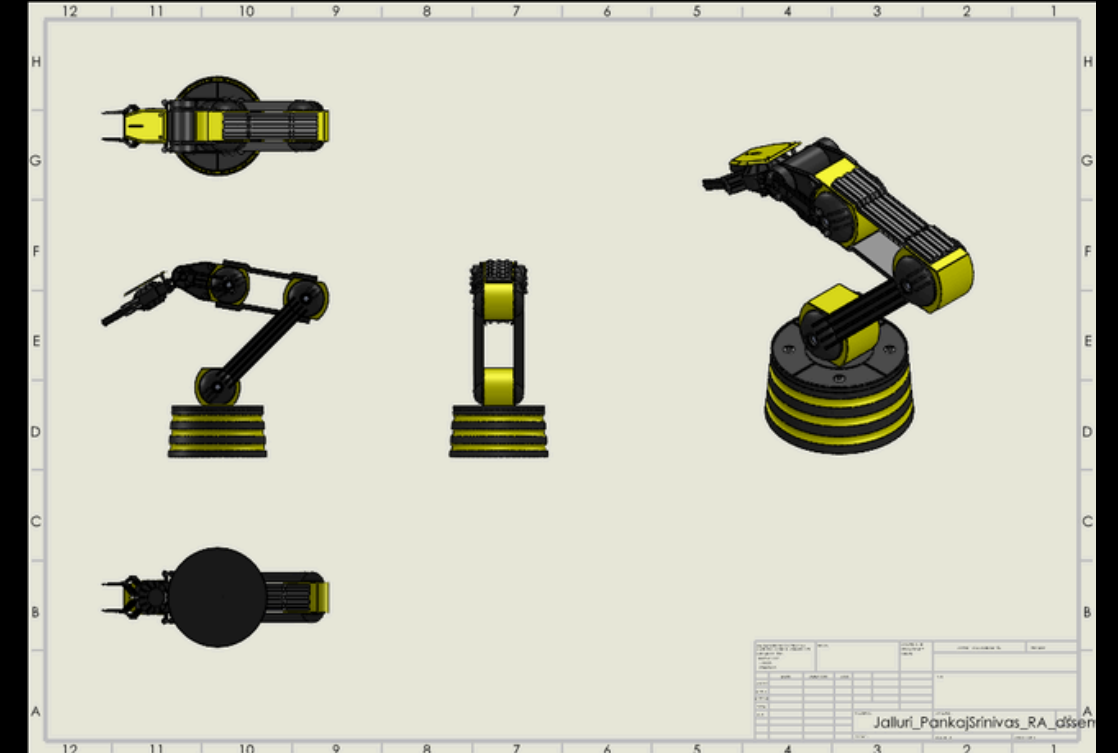
07



Instron Test Fixtures (MSA)



ME 1670 Final Project - OmniTrek



Robotic Arm • Douglas DC-3